

Universe24: the group of order 24 behind the clicks, one local integer operator on the cube's lattice, and the experiments it confronts

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Abstract

Can one local integer operator on a cubic lattice, read only through detectors' clicks, confront the quantum and relativistic experiments? The whole GameBoard is one element of one integer module, and the law is one linear operator on it: at every Node a second-order step on the Node's two levels and its six neighbours, with one integer pair per family and the division's remainder kept. The lattice's symmetries are the 48 signed permutations of the axes and their 24 rotations. The click is the only non-local act: when a record's one-way inward flux into a detector crosses its threshold, the record ends whole, at once. The birth, the other side of a click, is the only write. Exact on the lattice: the conserved quadratic forms, the step's bijection, the local flux identity, one click per quantum, Born's rule for a detector to the grain of the record's wheel, and no signalling through the click. A product state gives the Clauser–Horne–Shimony–Holt (CHSH) sum at most 2; an entangled pair on integer polariser axes gives 14/5. Recovered at long wavelength: $c = 1/\sqrt{3}$, isotropic at second order, and a moving packet's clock slowed by its own dispersion. The paper has twenty-four experiments: eighteen measured in clicks against pins written blind, ten with their pin, seven still owed and one a diagnostic, and six algebraic. None is run yet on this engine.

Keywords: integer lattice model; the symmetry group of the cube; one linear operator; Bell correlations; discrete relativity; pre-registered simulation

1 Introduction

The group was found, not assumed. The operations came first, as acts on the board, and the road is told in Section 7: a message crossed one Link per interval, messages met at a Node, a detector clicked. The group is the answer to which maps preserve what the acts work on, a Node with six Ports and its Links and a causal cone of one Link per interval: the bijections of the six Ports that keep opposite Ports opposite are the 48 signed permutations of the axes, and the only $\mathbb{Z}$ -linear bijections that preserve the cone are these 48, so no boost is among them and Lorentz's form is reached Outside from the clicks, not as a symmetry of the lattice. The 24 came from the hand, a pseudoscalar that a reflection flips and a rotation keeps, which splits the 48 into the 24 rotations and the 24 reflections. Every rule is required to commute with the 48, a requirement beyond locality. Today the law is one operator on one element, with no stepping body and no merge of messages [3, 9.19].

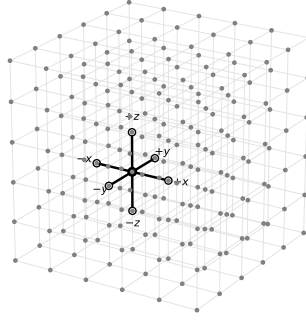
The whole move, in the order it is built. Seven things, each named once in the project's terminology [5].

- (1) *The GameBoard and its group.* The board is a cubic lattice of Nodes, six neighbours each. The symmetry group of the cube is the group of order 48: the 24 rotations, the group of order 24, and the reflections. The law is the same at every Node and commutes with the 48 and with the translations, and two things follow: the pace of light is the same in every direction at leading order, $c^2 = 1/3$ from the six reads, and the departure from it appears only in a short wave, in the fixed cubic pattern of the 48 (Eq. (16)).
- (2) *The one element.* The state of the whole board, every Node, every family and every label together, is one element $\mathbf{v}$ of one integer module: one summand per quantum, a *record*, with its two levels (now and one interval before) at every Node and label and the remainder of the one division [3, 9.19 (1)]. A *family* is an index of the summands: its vacuum pair [num, den] and its label module, and nothing else [3, 9.22 (1)]. A world has at most three: light, the experiment's matter and the holder of the tools' bodies.
- (3) *The one operator.* The law is one linear map $\mathbf{S}$ of that element (Eq. (1)): per family the second-order step with the family's pair at every Node, and at declared cells a coupling between two families and an integer label matrix [3, 9.19 (2)].
- (4) *The tools.* A tool is a region of the operator: a cube of the holder family with its material integers (a pair, a coupling, an axis), its total momentum $\mathbf{K}$ (the wave vector of its character, $\mathbf{0}$ at rest), at most a birth, and a name [3, 9.19 (5)]; no code belongs to a tool [4, A].
- (5) *The detector and the click.* A detector is a named region, a cube of side at least 3. The click is the only non-local act: a record's one-way inward flux into the detectors named on its ladder, summed over intervals, crosses the record's threshold, and the record ends whole, everywhere, at that interval; the click is the detector's and names no Node inside it [3, 9.25].
- (6) *The birth.* The only write, the other side of a click: the click of an emitter body's own excited record writes the born record's two levels once on the emitter's cells, the transpose $\mathbf{E}^T$ of the reading, and from then on only the law advances it [3, 9.13, 9.17].
- (7) *The experiment.* A list of entries on the board, each its family, how much and where, with its momentum; a generator writes the initial state once from the operator itself, the engine checks it in integers at load and runs the law, and the output is the clicks, compared with a pin written before the run (Section 2, "The experiment as data"; Section 6).

The general formula. The law is one step $\mathbf{S}$ of the one element at every interval, and two acts on its summands:

the step: $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M}_F \mathbf{a}_{\text{now}} + \mathbf{c}_F, \quad \mathbf{M}_F = \mathbf{D}_F^{-1} \mathbf{A}/3,$ per family F, with the remainder kept; the click: $2W C(t^*) \geq (2u + 1) T, \quad C(t) = \sum_{s \leq t} \sum_{j \rightarrow i} \max(G_{ij}(s), 0),$ the sum over the Ports $j \rightarrow i$ into the ladder's detectors; the birth: $(\mathbf{a}_{\text{now}}, \mathbf{a}_{\text{before}}) \leftarrow \mathbf{E}^T(\text{the born pair})$ on the emitter's cells, at the click's interval.	(1)
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In the step, $\mathbf{a}_{\text{now}}$, $\mathbf{a}_{\text{before}}$ and $\mathbf{a}_{\text{next}}$ are a record's levels over the Nodes now, one interval before and one interval on; $\mathbf{A}$ is the read matrix of the board (the six reads of every Node); $\mathbf{D}_F$ is the diagonal matrix of the per-Node weights $\text{den}_i/\text{num}_i$ of the family's pair; and $\mathbf{c}_F$ is the coupling's first difference of the other family's levels at the cells of a coupled body, with, at the cells of a polar body, the integer label matrix on the record's two label rows. In integers the step is Eq. (2), one division by 3den_i with the remainder kept. In the click, G_{ij} is the flux into Node i from Node j



one Node and its six Ports, the Links to its six neighbours

Fig. 1 The GameBoard in space: a cubic array of Nodes, here $6 \times 6 \times 6$, each joined to its six neighbours by Links through the Ports $\pm x$, $\pm y$, $\pm z$ (the faint lines); one Node and its six neighbours marked, the Nodes one interval away from it, whose convex hull is the octahedron of Figure 2. Each record keeps at a Node its two levels and its remainder there, and a body's region keeps its material integers; nothing else sits at a Node. Drawn from the definitions; no run.

(Eq. (9)), T the record's conserved form, u its residue and W its wheel, both read from the law at its birth (Definition 1); the click falls at the first interval t^* at which the inequality holds, at the detector whose share of that interval's flux holds the threshold, and deletes the record. The birth is fired only by a click. Nothing else happens on the board: no table, no rate, no drive, no take and no declared residue. Section 2 states each piece with its proof or its mark; every derivation of the paper names the piece it starts from.

Physics as the passage of information. What the law describes is the passage of information from an emitter to a detector, and nothing between them is read. A record is born at an emitter's click with its two levels written once; it moves on the lattice by the one operator, at most one Link per interval; it leaves at a click, where the record ends whole, and the click is what is read: reality Outside is the clicks, and a number of the board is a diagnostic. Transport is local at every step; the one non-local act is the click itself, the record ending everywhere at once, and no information passes through it (Theorem 7). The road by which this object was found is Section 7; the engine on which it is computed is Section 6.

The reading rule and the notation. Only a detector's click is a measurement, and an experiment is compared with nature and with its pin through its receivers' clicks alone; the operator, the tools, conservation and the initial state are proved in the algebra and checked by the property test and the load check, never by an experiment's run [3, 9.21 (4)]. Every physics name in this paper (Born's, Tsirelson's, Malus's, Lorentz's) names the known form that the law's result recovers. A number of the GameBoard's state is a diagnostic of the host, labelled so, never compared with nature and never pinned. Every symbol is named in English at its first use, a scalar plain, a vector in bold lowercase, a matrix or an operator in bold uppercase.

What this paper contributes, and what it does not claim. New here: one explicitly stated integer operator with its click and its birth, whose conserved forms, bijection, flux identity, Born's rule and no-signalling are theorems on the lattice; experiments as short lists of entries, their initial states generated from the operator itself and checked in integers at load; and twenty-four experiments: eighteen measured, placed as tools, ten with a pin written blind before any run and seven with their pin still owed (Table 1), and six algebraic, each with how it is computed (Table 2).

Not claimed: a derivation of the Lorentz group, which is not among the 48; a moving rigid rod, which the model does not yet define (it has clocks and no rods, [3, 9.24 (6)]); or any comparison with nature before the experiments are run on this engine. Gravity is not measured in this paper: its forms under a shell average appear only as an algebraic experiment, a limit of the algebra and not a run.

2 The algebra: one element, one operator, the click and the birth

This section states the object and nothing else: the board and its group, the one element, the one operator with its conserved forms, the flux, the click, the birth, the families and the experiment as data. Each statement carries the mark of the algebra document [3]: PROVED (a proof for every board, in the document or here), DERIVED (from proved statements, by argument), COMPUTED (a computation, not a proof), or CARRIED (a design's proof, cited).

The definitions, in order. Five objects, each one line of mathematics before its words.

- (1) The GameBoard: the Nodes of a box in $\mathbb{Z}^3$ whose axes are periodic or open; the six Ports of a Node, $P = \{\pm x, \pm y, \pm z\}$, with the involution $e \mapsto -e$ (the opposite Port); at most one Link per interval.
- (2) The group of order 48 and its rotations: $G_{48} = \{g : P \rightarrow P \text{ a bijection with } g(-e) = -g(e)\}$, the 48 signed permutations of the three axes; $\det : G_{48} \rightarrow \{+1, -1\}$ a homomorphism; $G_{24} = \ker \det$, the rotations, the group of order 24 (Theorem 1). The hand, a pseudoscalar, $h \mapsto \det(g) h$, tells the two cosets apart.
- (3) The one element: $V = \bigoplus_q V_q$, $V_q = (\mathbb{Z}^{\text{Nodes}} \otimes L_q)^2 \otimes (\text{the remainders})$, one summand per quantum q , L_q its label module ($\mathbb{Z}^2$ per arm; $\mathbb{Z}^2 \otimes \mathbb{Z}^2$ for a record of rank 2); the state of the world at an interval is one element $\mathbf{v} \in V$ [3, 9.19 (1)].
- (4) The one operator: $\mathbf{S} : V \rightarrow V$, per summand of family F the step of Eq. (1), plus the coupling between summands of coupled families and the label matrix on the axes' cells; linear, local and equivariant [3, 9.19 (2), (6)].
- (5) The two acts: the click, a record's one-way inward flux $\mathbf{E}$ against its form, the rung and the deletion of the summand, the only non-local act; and the birth $\mathbf{E}^T$, the only write, at the click's interval (Definitions 1 and 2).

The words of this paper. The *GameBoard* is the cubic array of Nodes, each joined to six neighbours by Links through directional Ports; time is a sequence of intervals. *Inside* is the board and what moves on it, where no one measures; *Outside* is the game above the board, the detectors and their clicks, the only place a number is read. A *record* is one summand of the element, one quantum: its two levels over the Nodes and its labels, with the remainder kept per Node on the record's own row. A *family* is an index of the summands, its vacuum pair and its label module. A *body* is a region of the operator where a family's pair differs from the vacuum's; a *tool* is a region with its integers, at most a birth and a name; a *face* is the operator's zero row beyond an open border; a *receiver* or *detector* is a name the host gives a region, on records' ladders [3, 9.19 (5)]. Every number carries its *kind*: DETECTOR, a click or a count of clicks, the only measurement; GAMEBOARD, a number of the board's state, a diagnostic and never a result; COMPUTATION, a number computed from the algebra before any run; DECLARATION, a declared integer of an experiment; HOST, a number of the host's bookkeeping, never a reading. Figure 1 draws the board in space.

The six Ports, the group and the front. Everything geometric in the law follows from one Node and its six Ports, $\pm x, \pm y, \pm z$, and from at most one Link per interval. The maps of the six

Ports onto themselves that keep opposite Ports opposite are the signed permutations of the three axes, 48 in all, the symmetry group of the cube and of the octahedron. The determinant splits them into two halves of 24: the rotations, which permute the cube's four body diagonals, and the improper maps, the reflections with the central inversion among them.

Theorem 1 (the 24 of the name). *The maps of the six Ports onto themselves that send opposite Ports to opposite Ports form the group of the signed permutations of the three axes, of order $2^3 \cdot 3! = 48$. The determinant is a homomorphism of this group onto $\{+1, -1\}$; its kernel, the group of order 24 (the rotation group of the cube), has index 2 and is isomorphic to the symmetric group S_4 of the permutations of the cube's four body diagonals; the other coset, the reflections, has 24 elements too. The hand, a pseudoscalar that a map g sends to $\det(g)$ times itself, tells the 24 rotations from the 24 reflections, as does an axial vector.*

The proof is in Appendix A.

The rule per family. At every Node i a record's row $(a_{\text{now}}, a_{\text{before}}, r)$ is mapped once per interval by

$$3 \text{den}_i a_{\text{next}} + r' = \text{num}_i S_6 - 3 \text{den}_i a_{\text{before}} + r, \quad 0 \leq r' < 3 \text{den}_i, \quad (2)$$

then $a_{\text{before}} \leftarrow a_{\text{now}}$, $a_{\text{now}} \leftarrow a_{\text{next}}$, $r \leftarrow r' [3, 8.1]$. Here S_6 is the sum of the six reads of the record's a_{now} (the neighbour across a Link; the Node itself on a folded axis of extent 1; 0 beyond an open face), $[\text{num}_i, \text{den}_i]$ the family's pair at the Node (the vacuum's pair everywhere, a body's pair on its cells) and r the remainder of the division, kept on the record's row. Light is the pair $[1, 1]$; a massive family is a pair with $\text{den} > \text{num}$, and nothing else names the mass. The step is made of three verbs on the row: the bilinear form (B), one entry num_i on the six reads and one entry -3den_i on the row's past; the translation (T) of the row by that rate; and the division (D) by the wall 3den_i with the remainder kept. The three tests hold [3, 8.1]: generic, one primitive with one declared pair and no family name; vector, the three verbs with no root and no float; local, the record's own row and its six reads, nothing kept at a Node beyond the row and its remainder.

The conserved form. Let $\mathbf{A}$ be the read matrix of the board, A_{ij} the number of reads of j among the six reads of i (the neighbour across a Link once; the Node itself once per folded direction; nothing beyond an open face). It is symmetric, because i reads j in the direction d exactly when j reads i in $-d$, and every row sums to at most 6. Let $\mathbf{D} = \text{diag}(D_i)$ with $D_i = \text{den}_i / \text{num}_i \geq 1$ and $\mathbf{M} = \mathbf{D}^{-1} \mathbf{A} / 3$. Dividing Eq. (2) by 3den_i , the level is

$$\mathbf{a}_{\text{next}} = \mathbf{M} \mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}} + \mathbf{e}, \quad e_i^{\text{rem}} = \frac{r_i - r'_i}{3 \text{den}_i}, \quad (3)$$

the rule without the remainder being $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$ ($\mathbf{e} = \mathbf{0}$), and $\mathbf{e}$ the vector of the remainder terms e_i^{rem} .

Theorem 2 (the conserved form). *Without the remainder, the form*

$$\begin{aligned} I(\mathbf{a}_{\text{next}}, \mathbf{a}_{\text{now}}) &= \mathbf{a}_{\text{next}} \cdot \mathbf{D} \mathbf{a}_{\text{next}} + \mathbf{a}_{\text{now}} \cdot \mathbf{D} \mathbf{a}_{\text{now}} - \mathbf{a}_{\text{next}} \cdot \mathbf{D} \mathbf{M} \mathbf{a}_{\text{now}} \\ &= \sum_i D_i (a_{\text{next},i}^2 + a_{\text{now},i}^2) - \frac{1}{3} \sum_i a_{\text{next},i} (\mathbf{A} \mathbf{a}_{\text{now}})_i \end{aligned} \quad (4)$$

is conserved on every board and every extent. With the remainders kept, the change in one interval is exactly

$$I(t) - I(t-1) = \sum_i D_i e_i^{\text{rem}} (a_{\text{next},i} - a_{\text{before},i}) = \sum_i \frac{(a_{\text{next},i} - a_{\text{before},i})(r_i - r'_i)}{3 \text{num}_i}, \quad (5)$$

an identity in the state the engine holds, with no tolerance.

The proof is in Appendix A. In the engine's units, times 3 num for one pair, the form is $3 \text{den} (a_{\text{next}}^2 + a_{\text{now}}^2)$ summed over the Nodes less num $(a_{\text{next},i} a_{\text{now},j} + a_{\text{next},j} a_{\text{now},i})$ summed over the Links, an integer, and the change of Eq. (5) is the integer $\sum_i (a_{\text{next},i} - a_{\text{before},i})(r_i - r'_i)$, bounded by 3 den times the motion. A static level ($\mathbf{a}_{\text{now}} = \mathbf{a}_{\text{before}}$ everywhere) has $I = 0$, so it clicks nowhere; the sum of the squares would count it.

Proposition 1 (positive definiteness). *I is positive definite for every massive pair ($\text{den}_i > \text{num}_i$ at every Node) on every board, and semidefinite at light's pair.*

The proof is in Appendix A.

The modes and the stability condition. Restricted to one family without the remainder, $\mathbf{S}$ is the second-order linear recurrence $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M}\mathbf{a}_{\text{now}}$. In the variables $\mathbf{D}^{1/2}\mathbf{a}$ it reads with the symmetric $\mathbf{N}$, whose orthonormal eigenvectors $\mathbf{q}$ are the board's *modes*: on a mode with the eigenvalue $\lambda = 2 \cos \omega$ the recurrence $q(t+1) + q(t-1) = 2 \cos \omega q(t)$ has the solutions $\cos(\omega t)$ and $\sin(\omega t)$, a rotation at the angle ω per interval, the mode's clock, while an eigenvalue $|\lambda| \geq 2$ gives cosh and sinh, a growth without bound. On a torus there is one mode per Node. A mode is *bound* when $2 \cos \omega$ lies above the vacuum band's top and *free* inside the band (Section 3).

Theorem 3 (the stability condition). *Every mode is a rotation, and I is positive definite, exactly when the largest eigenvalue of $\mathbf{N}$ is below 2.*

Proof. By Proposition 1's block form, I is positive definite exactly when every $|\lambda| < 2$. The matrix $\mathbf{N}$ has nonnegative entries, so by Perron–Frobenius its largest eigenvalue λ_{max} is the largest in absolute value, and every eigenvalue lies in $[-\lambda_{\text{max}}, \lambda_{\text{max}}]$. So $\lambda_{\text{max}} < 2$ puts every eigenvalue in $(-2, 2)$: the one check bounds both ends [3, 9.19 (2)]. $\square$

A pair with $\text{num} \leq \text{den}$ at every Node is sufficient (Proposition 1), not necessary: a one-cell well [8, 7] on the vacuum [7, 8] has $\lambda_{\text{max}} = 1.90$ on a chain and is stable, while [800, 700] on [800, 809] has 2.03 and runs away (COMPUTATIONS of the algebra, 9.19 (2)). The loader's check is the spectral one: the composed operator's largest eigenvalue below 2, the file refused otherwise. Light cannot be a body: an index for light has $D_i \geq 1$ everywhere, so $\|\mathbf{N}\| \leq 6/3 = 2$, no eigenvalue lies above the massless band's top and no bound mode of light exists [3, 8.3].

The coupling of two families. At a cell of a coupled body R the two records' rows are coupled by one entry of verb (B) over the other record's two levels, the first difference both ways, with g and G the two declared coupling pairs at the body's cells (the capital G apart from the flux G_{ij} below, which always carries two Node indices; CARRIED [3, 8.5]):

$$\begin{aligned} \mathbf{a}_{m,\text{next}} + \mathbf{a}_{m,\text{before}} &= \mathbf{M}\mathbf{a}_{m,\text{now}} + g \mathbf{P}_R(\mathbf{a}_{l,\text{now}} - \mathbf{a}_{l,\text{before}}), \\ \mathbf{a}_{l,\text{next}} + \mathbf{a}_{l,\text{before}} &= \frac{1}{3} \mathbf{A}\mathbf{a}_{l,\text{now}} - G \mathbf{P}_R(\mathbf{a}_{m,\text{next}} - \mathbf{a}_{m,\text{now}}), \end{aligned} \tag{6}$$

the massive record m receiving light's backward difference and light l receiving the massive record's forward difference as its source, $\mathbf{P}_R$ the projector onto the cells (1 on R , 0 elsewhere); the massive step is taken first and light's step reads the massive level just written. The body declares one pair on its coupled cells, so $\mathbf{P}_R \mathbf{D} = D_{\text{in}} \mathbf{P}_R$, and we write $\alpha = g D_{\text{in}} / G$, one number of the body.

Proposition 2 (the scheme is variational). *Eq. (6) is the discrete Euler–Lagrange equation of the two-point Lagrangian*

$$L(t) = -\mathbf{a}_{m,t} \cdot \mathbf{D}\mathbf{a}_{m,t+1} + \frac{1}{2} \mathbf{a}_{m,t} \cdot \frac{1}{3} \mathbf{A}\mathbf{a}_{m,t} + \alpha (-\mathbf{a}_{l,t} \cdot \mathbf{a}_{l,t+1} + \frac{1}{2} \mathbf{a}_{l,t} \cdot \frac{1}{3} \mathbf{A}\mathbf{a}_{l,t}) + g \mathbf{a}_{m,t+1} \cdot \mathbf{D}\mathbf{P}_R(\mathbf{a}_{l,t+1} - \mathbf{a}_{l,t}).$$

The proof is in Appendix A. Being variational and linear, the scheme is symplectic, and a symplectic linear map has an exact quadratic invariant.

Theorem 4 (the coupled invariant). *With I_m the massive form of Eq. (4) and I_l light's form with the weight 1,*

$$J = I_m + \alpha I_l + g \sum_{i \in R} D_i (a_{m,\text{next}} - a_{m,\text{now}})_i (a_{l,\text{next}} - a_{l,\text{now}})_i \quad (7)$$

is conserved exactly on every board and every extent, before the remainders.

The proof is in Appendix A. The amplitude form of the coupling, light's row gaining g times the massive level, is withdrawn as tachyonic: its mass matrix at zero wave vector, $\begin{pmatrix} 0 & -g \\ -g & \omega_0^2 \end{pmatrix}$, has the determinant $-g^2 < 0$ and so one negative eigenvalue [3, 8.5].

The label matrix. Every record carries its rows per label value of each arm. A polar body with the integer axis $\mathbf{u} = (a, b)$ applies at its cells the integer matrix $\mathbf{M}_{\mathbf{u}} = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$ to the record's pair of label rows (x, y) , verb (B), with no division. Its rows are orthogonal, each of squared length $a^2 + b^2$, so $\mathbf{M}_{\mathbf{u}}^T \mathbf{M}_{\mathbf{u}} = (a^2 + b^2) \mathbf{1}$, which is Brahmagupta's identity

$$(ax + by)^2 + (-bx + ay)^2 = (a^2 + b^2)(x^2 + y^2), \quad (8)$$

exact on integers: the common factor $a^2 + b^2$ scales the record's form and cancels in every share [3, 9.16 (2)]. The polariser's angle is this integer axis, $\arctan(b/a)$ on the comparison side only; there is no table of angles [3, 9.14 (b)].

The flux. Let $e_i = D_i (a_{\text{next},i}^2 + a_{\text{now},i}^2) - \frac{1}{3} a_{\text{next},i} (\mathbf{A} \mathbf{a}_{\text{now}})_i$ be the Node's share of the form, so that $I = \sum_i e_i$.

Proposition 3 (the local flux identity). *Without the remainder, between two intervals,*

$$e_i(t) - e_i(t-1) = \sum_{j \text{ read by } i} G_{ij}, \quad G_{ij} = \frac{1}{3} A_{ij} (a_{\text{now},i} a_{\text{before},j} - a_{\text{before},i} a_{\text{now},j}), \quad G_{ij} = -G_{ji}. \quad (9)$$

The proof is in Appendix A. G_{ij} is the flux into Node i from Node j : a bilinear form of the record's own two levels at the two ends of a Link, free of the pairs, antisymmetric and local. For a character $A \cos(\omega t - \mathbf{k} \cdot \mathbf{x})$ moving from j to i along the Link's direction it is positive: with $\theta = \omega t - \mathbf{k} \cdot \mathbf{x}_j$ and $\mathbf{x}_i = \mathbf{x}_j + \hat{\mathbf{n}}$ one Link on, $a_{\text{now},i} a_{\text{before},j} - a_{\text{before},i} a_{\text{now},j} = A^2 [\cos(\theta - k) \cos(\theta - \omega) - \cos(\theta - k - \omega) \cos \theta] = \frac{1}{2} A^2 [\cos(\omega - k) - \cos(\omega + k)] = A^2 \sin \omega \sin k$, so $G_{ij} = \frac{1}{3} A^2 \sin \omega \sin k$, k the wave number along the Link, independent of the time. The one-way inward flux into one Node, summed over intervals and Ports of $\max(G_{ij}, 0)$, equals the record's form over the passage of a packet (COMPUTED to six digits on a chain, 2.135254 against 2.135254 [3, 9.19 (3)]). The rectification $\max(G, 0)$ is the click's own comparison: a record's energy is counted where it enters.

The residue and the wheel. Two things are called a remainder. The rule's remainder r is kept per Node on the record's row and is the law's own. A record's residue u is one integer, read once: the rule's remainder at the record's birth cell at the interval of the click that births it; its wheel is $W = 3 \text{ den} / \text{gcd}(\text{num}, 3)$ of that cell's pair for the clicking record's family, in lowest terms. No residue, seed, order or wheel is declared anywhere; the loader refuses them (ADOPTED, record 1872 [2]; [3, 9.19 (4)]). The range of the remainder is fixed by the pair:

Proposition 4 (the remainder's range). *With the pair in lowest terms and every remainder 0 at interval 0, every remainder at a Node is a multiple of $\gcd(\text{num}, 3)$, so at most $3 \text{ den} / \gcd(\text{num}, 3)$ values occur.*

Proof. By Eq. (2), $r' \equiv \text{num } S_6 - 3 \text{ den } a_{\text{before}} + r \equiv \text{num } S_6 + r \pmod{3 \text{ den}}$. If r is a multiple of $g = \gcd(\text{num}, 3 \text{ den})$, so is $\text{num } S_6 + r$, and so is its residue modulo 3 den , a multiple of g ; by induction from $r = 0$ every remainder is. Since $\gcd(\text{num}, \text{den}) = 1$, $g = \gcd(\text{num}, 3)$, and the multiples of g in $[0, 3 \text{ den})$ number $3 \text{ den} / g$ [3, 9.19 (4)]. $\square$

That every one of these values is reached, and equidistributed, is COMPUTED, not proved: on a chain the vacuum [800, 809] gives 2427 values equidistributed over 18000 intervals (the chi-square 21 on 19 degrees of freedom), and a pair [1, 1] gives three [3, 9.19 (4)]. The residues of an emitter spread only by the board's own back-action: a body reseeded identically after every click, with no coupling to the light on the board, repeats one residue, a perfectly periodic source; so every emitting body declares its coupling (g, G) to the family it births, and the excited record's residue is read at its centre cell after its first advance [3, 9.19 (4e)]. The granularity of Born's rule on a wheel of W values biases a share by at most $1/(2W)$, so every birth cell carries a pair with $W \geq 500$, checked at load [3, 9.19 (4a)]. With the law's remainder the residues are equidistributed and not a permutation, and the counts are binomial about the shares, as nature's are; the run stays deterministic and reproducible.

Definition 1 (The detector and the click). A *detector* is one connected named region, a cube of side at least 3 (a smaller cube, or one name on disconnected pieces, is refused at load); separate places are separate names [3, 9.25 (7)]; record 1899 [2]. A record carries a *ladder*, the detectors it names in a declared order with the face receiver last. At its birth the record's threshold is fixed, $\theta = (2u + 1)T/(2W)$, with T the record's form I , and its running total starts at $C = 0$. At every interval t each detector d of the ladder receives the increment

$$f_d(t) = \sum_{\substack{i \in d, j \notin d \\ j \text{ read by } i}} \max(G_{ij}(t), 0), \quad (10)$$

the one-way inward flux through the region's Ports from outside it (a Link between two Nodes of one detector is no Port of it), and $C(t) = C(t-1) + \sum_d f_d(t)$. The *click* falls at the first interval t^* with $C(t^*) \geq \theta$, that is $2W C(t^*) \geq (2u + 1)T$, at the detector whose segment of that interval's increment, laid out in the ladder's order, contains $\theta - C(t^* - 1)$; there the record ends, whole, everywhere, at that interval. The click is the detector's: it names the detector, never a Node inside it, and is reported per detector [3, 9.25 (2)]; record 1899 [2]. T and C are integers once scaled by 3 and by the pairs' common wall.

The three tests hold for the click: generic, the integers u , T , W and the fluxes, no family name; vector, sums and one comparison per detector, the division's comparison (D), no root; local, each detector's increment is its own Ports', and the record's two integers θ and $C(t-1)$ travel with the record as its residue does. What is not local is the deletion: the record ends at once, everywhere, the one non-local act of the law (the model owner's decision, record 1888 [2]).

Theorem 5 (one quantum, one click). *With the face receiver last on every ladder, every record clicks exactly once.*

The proof is in Appendix A.

Theorem 6 (Born’s rule for a detector). *Let the residue u be spread evenly over $\{0, \dots, W - 1\}$. Then the probability that a record clicks at detector d is $C_d(\infty)/T$, the detector’s share of the record’s total inward flux, whatever the flux’s time profile, to the grain $1/W$; over the ladder with the face the shares sum to one. A detector’s share is the sum of its Nodes’ shares.*

The proof is in Appendix A. The residues of the law’s remainder are equidistributed (COMPUTED), not a permutation, so the counts are binomial about the shares [3, 9.19 (4b)]. Two other forms were weighed and refused: a race in which each Node shouts on its own threshold is not Born’s rule (two Nodes receiving the shares 0.9 and 0.1 give 17/18 to the first, PROVED), and exactness needs the chance normalised by the record’s remaining norm, which is the increment ladder read cell by cell [3, 9.25 (5)].

Definition 2 (The birth). The reading $\mathbf{E}$ maps a record at a set of cells to its offer there; its transpose $\mathbf{E}^\top$ maps a number back to a record, the real part of a character written on cells [3, 9.13]. A birth is $\mathbf{E}^\top$ at a click, and only at a click. An *emitter* is a body of a massive family with a stock of M excitations: at interval 0 its first excited record is on the board, the body’s mode at both levels; excited record k clicks at its own rung on its own centre cell, and at that click it ends and $\mathbf{E}^\top$ writes two records, the born record (one quantum) and, while the stock lasts, excited record $k + 1$ (one quantum); so quanta are conserved at every click, and the emission times are the rungs, the spontaneous emission of the algebra [3, 9.17 (2), (4)]. The born record’s two levels are integers of the world written by the generator, on one cell the pair [now, –now], on a line of cells a travelling profile, copied once at the click and never driven afterwards [3, 9.17 (5), (6)]. A *crystal* is $\mathbf{E}$, the arriving record’s click at its cells, then the split of the arriving clock into the two born clocks, then $\mathbf{E}^\top$, the birth of an entangled pair with its declared branches [3, 9.13]; record 1879 [2].

Where a tensor enters. In one place only: the pair, a record of fixed rank 2 whose label module is $\mathbb{Z}^2 \otimes \mathbb{Z}^2$, four integer weights on the joint labels, each arm’s rows over the Nodes as any record’s. It is required only where entanglement is measured, since a product state gives the CHSH sum at most 2 (PROVED: CHSH’s bound holds for weights that factor). It is born only by the crystal during the run, never written at interval 0, and two separately born records are never joined; so the generator writes only vectors. The pair’s click reads its joint weights $R(o_A, o_B) = J(o_A, o_B)^2$, $J(o_A, o_B)$ the joint amplitude of the outcomes o_A and o_B (apart from the invariant J of Eq. (7)), on the joint ladder of the four outcomes, the one residue choosing the outcome for both arms at once; the click of one arm alone reads the weights summed over the other arm’s labels, the partial trace, kept nowhere [3, 9.22 (6), 9.25 (6)]; record 1880 [2].

Theorem 7 (no signalling through the click). *For a pair record, the share of records clicking + at Alice’s detector is $|\mathbf{w}_+|^2 / (|\mathbf{w}_+|^2 + |\mathbf{w}_-|^2)$, with $\mathbf{w}_o$ her arm’s label state projected on her own axis, exactly in integers and independent of Bob’s axis, placement and distance; the same holds for Bob. So the counts at one detector do not depend on what is done at the other, and no information passes through the click, at any speed.*

The proof is in Appendix A. What is non-local is Bob’s outcome given Alice’s, decided by the same residue and the joint weights: the correlation, not a signal. A deletion that spread from Alice’s click at one Link per interval would be a local rule at each arm, and every such rule gives the CHSH sum at most 2; so the deletion is at once, the algebra’s one non-local act [3, 9.25 (6)]; record 1888 [2].

Theorem 8 (the direction of time). *Given the neighbours’ reads, the step $(a_{\text{before}}, r) \mapsto (a_{\text{next}}, r')$ of one row is a bijection of $\mathbb{Z} \times \{0, \dots, 3 \text{ den} - 1\}$ onto itself, with the exact inverse $a_{\text{before}} = \lceil m / (3 \text{ den}) \rceil$,*

$r = 3 \text{ den } a_{\text{before}} - m$, where $m = \text{num } S_6 - 3 \text{ den } a_{\text{next}} - r'$. The coupled step is two such bijections in a fixed order, inverted in the reverse order. The click deletes a summand and has no inverse.

The proof is in Appendix A. The Inside is reversible to the bit on every board and is not symmetric in form, the remainder living on the later level: the rule carries an arrow in its bookkeeping and none in its information. The arrow of time is born at the click and nowhere in the rule. The check that shows it is a property test and never a pin: a planted board with no detector run forward and back under the exact inverse returns bit for bit, and the same board with a detector does not, by the deleted record exactly [3, 9.20].

The families. A family is its vacuum pair $[\text{num}, \text{den}]$ and its label module ($\mathbb{Z}^2$ per arm for the hand, $\mathbb{Z}$ for a family with no hand), and nothing else [3, 9.22 (1)]. From the pair alone follow its rest frequency, $\cos \omega_0 = \text{num}/\text{den}$ (light has none), its operator block on an empty board and its band (Section 3), its stability, whether it can hold a body (only a massive family can), and its remainder's range. A born record's clock is a datum of the birth that makes it, not of the family. A world has at most three families, fixed by one pair per family per Node: light, the experiment's matter and the holder of the tools' bodies (ADOPTED, record 1875 [2]).

The experiment as data. An experiment is a list of what is on the board, and nothing else [3, 9.22 (2)]; records 1875, 1882 and 1885 [2]:

- (i) the board: the extents per axis, and per axis periodic or open, an open axis carrying the face receiver at its border, one border for every family;
 - (ii) the families, at most three, each its pair and label module;
 - (iii) the material map: which Nodes carry which pair per family, which a coupling (g, G) , which an axis (a, b) , every region a cube by its vertex and edge;
 - (iv) the occupation: which bound bodies hold quanta and how many, at which amplitude; which packets are on the board; which bodies carry a stock and birth which family with which born profile; which body births a pair, with the crystal's branches;
 - (v) the momentum: every entry carries its total momentum, the wave vector $\mathbf{K}$ of its character, $\mathbf{0}$ at rest; there is no acceleration, a moving entry is written moving at interval 0 and its energy of motion is part of the initial state (record 1884 [2]);
 - (vi) the names: the detectors' regions and each birth's ladder, the face receiver last.
- Not part of it, and refused at load: a residue, a wheel, a seed written by hand, a rate, a train, a heading, a take, a grace, a table, a per-family border, a ramp.

The initial state. The initial element is

$$\begin{aligned} \mathbf{v}_0 = & \sum_{\text{occupied bound modes}} (\text{the quanta}) \times (\text{the mode's integer profile, at both levels}) \\ & + \sum_{\text{packets}} (\text{the profile}) \times (\text{the character of } \mathbf{K}), \end{aligned} \tag{11}$$

each profile at the declared amplitude, with every remainder 0 and nothing else on the board: no light record and no remainder at interval 0 [3, 9.22 (3)]. The modes are those of the composed operator of the body's family, every region in place. The generator is the board's own operator iterated.

Proposition 5 (the generator relaxes with the law's read). *The power iteration $\mathbf{v} \mapsto (\mathbf{M} + 2\mathbf{1})\mathbf{v}$, $\mathbf{1}$ the identity, converges from any start with a nonzero overlap to the bound mode of the composed operator, the eigenvector of its largest eigenvalue λ , at the rate $1 - \text{gap}/(\lambda + 2)$ per iteration, gap the distance of λ from the next eigenvalue.*

The proof is in Appendix A. The iteration applies the law's own read (the six neighbours with the pairs at every Node) and nothing else. In integers, $3 \text{ den } \mathbf{v}' = \text{num } S_6(\mathbf{v}) + 6 \text{ den } \mathbf{v}$ with the remainder kept and an exact shift by a power of two when the levels pass the amplitude, it is reproducible bit for bit on any host; the float Lanczos iteration is a host shortcut only [3, 9.22 (7)]. So the law *steps* the operator, the generator *relaxes* with it, and the click compares and deletes; nothing on the board relaxes.

The load check. The initial element is written once into the experiment's input file, with each occupied mode's clock as a rational a/b , and the engine checks it at load in integers only and says LAWFUL or REFUSED with the reason: the law and generator the file was made under (a change of the laws forces regeneration); the eigen-equation's residual at every Node of the profile $\mathbf{p}$,

$$|b \text{ num}_i (S_6 \mathbf{p})_i - 3 \text{ den}_i a p_i| \leq b (3 \text{ num}_i + 6 \text{ den}_i), \quad (12)$$

the bound for the rounded profile of an exact mode (Proposition 6); a/b above the band's top and below 2; every remainder 0; the stability of the composed operator; the rich birth cells; the detectors' cubes; and the refused keys [3, 9.22 (3), (7)]; record 1886 [2]. Recomputation at load is a diagnostic only, since a host's float mode computation is not reproducible across hosts.

Proposition 6 (the residual bound). *Let $\mathbf{q}$ be an exact mode, $\text{num}_i(S_6 \mathbf{q})_i = 3 \text{ den}_i \lambda q_i$ at every Node with $|\lambda| < 2$ and $|q_i| \leq A$, the amplitude. Let $\mathbf{p}$ be $\mathbf{q}$ rounded to the nearest integers, and a/b be λ rounded to the nearest multiple of $1/b$ with $b \geq A$. Then Eq. (12) holds at every Node.*

The proof is in Appendix A. A profile that is not a mode fails the bound by a margin of the amplitude's order; the check does not single out the mode among near-degenerate ones, whose hybrid share stays the generator's, printed as a GameBoard diagnostic [3, 9.22 (7)].

3 The massive family: the band, the cone, the bound body and the moving packet

The rule of Eq. (2) gives every family one declared pair and lets the one operator carry mass, a clock and a body [3, chapter 8, 9.24]. This section states what follows from the pair before any run; each sentence carries the algebra document's mark, and every number is a COMPUTATION declared before its run.

The band and the pitch. Read Eq. (2) without its remainder on a character $a(\mathbf{x}, t) = \cos(\omega t - \mathbf{k} \cdot \mathbf{x})$ of a periodic board, with ω the frequency angle per interval and $\mathbf{k}$ the wave vector (angles k_x, k_y, k_z per Link).

Theorem 9 (the band). *The character is a solution of the rule exactly when*

$$\boxed{\cos \omega = \cos \omega_0 \cos \omega_l(\mathbf{k}), \quad \cos \omega_0 = \text{num}/\text{den}, \quad \cos \omega_l(\mathbf{k}) = \frac{1}{3}(\cos k_x + \cos k_y + \cos k_z)}, \quad (13)$$

$\omega_l(\mathbf{k})$ being light's band at the same wave vector and ω_0 the family's pitch, its rest frequency. Every character has a real ω for every $\mathbf{k}$, the band lying in $[-\text{num}/\text{den}, \text{num}/\text{den}]$ in $\cos \omega$.

The proof is in Appendix A. At $\mathbf{k} = \mathbf{0}$ the gap is the pitch, the rest period $N_0 = 2\pi/\omega_0$ intervals and the rest energy $h\omega_0$, h the action; light has $\omega_0 = 0$, a zero mode that is a level and not a clock, so the whole difference between light and a massive family in the algebra is a gap. The band is exact on a periodic board for every extent; on a board with an open face the characters

are not eigenvectors and the statement is the periodic board's. With $\cos k_i = 1 - 2\sin^2(k_i/2)$ and $2 - 2\cos\omega = 4\sin^2(\omega/2)$ the band is, exactly,

$$4\sin^2\frac{\omega}{2} = 2\left(1 - \frac{\text{num}}{\text{den}}\right) + \frac{\text{num}}{\text{den}} \cdot \frac{4}{3}\left(\sin^2\frac{k_x}{2} + \sin^2\frac{k_y}{2} + \sin^2\frac{k_z}{2}\right), \quad (14)$$

light's form (num = den) with the pace squared scaled by num/den and a gap added [3, 8.1].

The two paces and the exact cone. Near $\mathbf{k} = \mathbf{0}$, $\cos\omega_l = 1 - k^2/6 + O(k^4)$ with $k = |\mathbf{k}|$, so $\omega_l^2 = k^2/3 + O(k^4)$: light's pace is $c = 1/\sqrt{3}$ Links per interval. For the massive band there are two second-order readings [3, 8.1]. (i) Jointly in ω and $\mathbf{k}$: $2 - 2\cos\omega = 2(1 - \cos\omega_0) + 2\cos\omega_0(1 - \cos\omega_l)$, and to second order $\omega^2 = \omega_0'^2 + \cos\omega_0 c^2 k^2$ with $\omega_0'^2 = 2(1 - \cos\omega_0)$: the Klein–Gordon dispersion with the cone

$$c_m^2 = \cos\omega_0 c^2, \quad c_m = c\sqrt{\cos\omega_0} = c(1 - \omega_0^2/4 + O(\omega_0^4)),$$

a massive record's limiting pace below light's by $\omega_0^2/4$ (0.56 percent at the pair [800, 809], $\omega_0 = \arccos(800/809) = 0.1493$, $N_0 = 42.08$: a COMPUTATION of the algebra, 8.1, not a pin). (ii) In $\mathbf{k}$ alone, with no truncation in ω_0 : write $\omega = \omega_0 + \delta$ at the band's bottom. Then $\cos(\omega_0 + \delta) = \cos\omega_0 - \sin\omega_0 \delta + O(\delta^2)$ and $\cos\omega_0 \cos\omega_l = \cos\omega_0(1 - \omega_l^2/2 + O(\omega_l^4))$, so $\sin\omega_0 \delta = \cos\omega_0 \omega_l^2/2$; and $\omega^2 = \omega_0^2 + 2\omega_0 \delta + O(\delta^2)$ gives

$$\boxed{\omega^2 = \omega_0^2 + c_{\text{eff}}^2 k^2 + O(k^4), \quad c_{\text{eff}}^2 = \cos\omega_0 \frac{\omega_0}{\sin\omega_0} c^2, \quad \frac{c_{\text{eff}}^2}{c_m^2} = \frac{\omega_0}{\sin\omega_0} = 1 + \frac{\omega_0^2}{6} + O(\omega_0^4)}, \quad (15)$$

c_{eff} the effective pace of the band's bottom, the named cone of the massive family (1.0037 for the ratio at $\omega_0 = 0.15$, a COMPUTATION of the algebra, 8.1, not a pin). The two readings agree at leading order in ω_0 and differ at order ω_0^2 .

Light against itself. Light's own band, the pair [1, 1], disperses by wavelength and by direction with nothing declared.

Proposition 7 (light's dispersion). *Along a unit direction $\mathbf{n}$ with $|\mathbf{k}| = k$, the group velocity $v_g = d\omega/dk$ of light's band is*

$$\boxed{\frac{v_g}{c} = 1 - \frac{(3\sum_i n_i^4 - 1)k^2}{24} + O(k^4)}. \quad (16)$$

The proof is in Appendix A. The coefficient $3\sum_i n_i^4 - 1$ is 0 on a body diagonal ($n_i^4 = 1/9$), 1/2 on a face diagonal ($n_i^4 = 1/4, 1/4, 0$), 2 on an axis, and 4/5 averaged over the sphere (the average of n_x^4 is 1/5); the series agrees with the exact group pace, 0.992472 against 0.992500 on an axis at $k = 0.3$ (COMPUTATION [7]). It is a prediction of form: the quadratic dispersion of light anisotropic with the cubic pattern of the 48 and no free coefficient; nature today has bounds and no measured coefficient (the algebraic experiment A2, Section 8.2).

The bound body at rest. A body is no new object: a region R of the operator, a cube of Nodes on which the 48 act, where the massive family's pair is $[\text{num}', \text{den}']$ with $\text{num}'/\text{den}' > \text{num}/\text{den}$, a well [3, 8.3]. Its clock is the composed operator's bound mode, the angle ω_b below the rest frequency ω_0 , whose $2\cos\omega_b$ lies above the vacuum band's top; the mode's tail falls exponentially away from the well (PROVED), so the torus must be large enough that its wrapped tail stays below half a unit at the amplitude [3, 9.22 (9)]. A bound tool body is of the holder family, distinct from the

experiment's matter (record 1875 [2]). At interval 0 the body's mode is written by the generator at both levels, and the load check verifies it (Eq. (12)).

The moving body. The one operator is linear in the state and, away from the regions of material, commutes with every translation.

Theorem 10 (no self-bound body). *A linear translation-invariant map has the characters as its modes and nothing else stationary, so a localised stationary state needs a region of material that breaks the translations. There is no self-bound localised body in the one algebra: a moving body is either a packet of a family carried by the character of its momentum, or a region of material, which does not move.*

The proof is in Appendix A. On a homogeneous board every character is a mode with $3 \cos \omega = (\text{num/den}) \sum_i \cos k_i$, and a packet, a sum of characters, is an exact solution, each character advanced by its own clock; the written initial state, the profile times the character of $\mathbf{K}$ at both levels, moves from interval 0 with no ramp (PROVED [3, 9.24 (2)]). Its group velocity follows by differentiating $3 \cos \omega = (\text{num/den}) \sum_i \cos k_i$: $-3 \sin \omega d\omega = -(\text{num/den}) \sin k_i dk_i$, so

$$v_i = \frac{\partial \omega}{\partial k_i} = \frac{\text{num}}{\text{den}} \frac{\sin k_i}{3 \sin \omega}, \quad (17)$$

and at small $\mathbf{K}$ and ω , $\omega^2 = \omega_0^2 + k^2/3$ with $\omega_0^2 = 2(1 - \text{num/den})$, the relativistic band with $c = 1/\sqrt{3}$ and the mass ω_0 [3, 9.24 (2)]. The packet's phase is $\varphi(\mathbf{x}, t) = \mathbf{K} \cdot \mathbf{x} - \omega(\mathbf{K})t$; at its centre, $\mathbf{x} = \mathbf{x}_0 + \mathbf{v}t$ with $\mathbf{v} = \mathbf{v}_g(\mathbf{K})$, $d\varphi/dt = \mathbf{K} \cdot \mathbf{v} - \omega(\mathbf{K})$, so the phase turns by $\omega(\mathbf{K}) - \mathbf{K} \cdot \mathbf{v}$ per interval and

$$\boxed{\frac{f}{f_0} = \frac{\omega(\mathbf{K}) - \mathbf{K} \cdot \mathbf{v}}{\omega_0}}, \quad (18)$$

f_0 the rest tick and f the moving packet's own tick: time dilation from the dispersion, with no rule for it. For the relativistic band $\omega^2 = \omega_0^2 + c^2 K^2$ the group velocity is $v = c^2 K/\omega$, so $\omega - Kv = (\omega^2 - c^2 K^2)/\omega = \omega_0^2/\omega$; with $\omega = \gamma\omega_0$, $\gamma = 1/\sqrt{1 - v^2/c^2}$ the Lorentz factor (since $\omega^2(1 - v^2/c^2) = \omega^2 - c^2 K^2 = \omega_0^2$), the ratio is exactly $1/\gamma$. On the lattice band along an axis at $v = 1/3$, the tick ratio 0.8146 (the pin of [3, 9.22 (8)], rows 5, 11 and 12; [4, B]). A packet of width σ (Links) spreads on the time $2\sigma^2/\omega''$, ω'' the curvature of the dispersion at rest, so a moving clock is a large packet, about a hundred Links wide for the experiments' holds [3, 9.24 (2)]. A detector or an emitter carried by a moving body is a *moving name*: its cells translate at the body's own group velocity from its $\mathbf{K}$, with no declared rate or heading; the click reads the flux into the moving region and the birth writes on it (the model owner's decision, record 1889 [2]; [3, 9.24 (4)]). There is no acceleration during a run: a moving entry is written moving at interval 0 and its energy of motion is conserved with the rest (record 1884 [2]). On the board's own surface nothing contracts: the profile is as written. The model has clocks and no rods: a moving rigid arm is not yet defined, and a light clock moving across its arm reads the packet's own dilation; the longitudinal clock's factor is not claimed (record 1895 [2]; [3, 9.24 (6)]).

The coupling and the index. A body coupled to light by the pairs g and G (Eq. (7)) scatters light at its cells; a medium body at rest has the index n with $n^2 = 1 + Gg/(\omega_0^2 - \omega^2)$ in closed form, the design's, its ω_0 the medium's pitch and ω light's frequency; the medium is a bound body of the holder family, its pin re-derived blind on that body (record 1877 [2]; [3, 9.22 (8)], rows 14 and 15).

The push. A body's cells stepping by its momentum, with its stress the flux of momentum at its outer Ports, is CARRIED from its design and not proved in the one algebra; momentum at a body

is conserved by declaration, not by the algebra, and a push from the law's own values is OPEN [3, 8.11, 9.21 (6), (7a)]. Every result this paper states as proved holds without it.

Known formulas and predictions. Every experiment of Section 8 carries (K), a known-formula row, whose algebraic identity is a textbook formula compared with, or (P), a prediction, a number the algebra gives and no known formula gives; a prediction is not a pass. The predictions of the massive family, each with its falsifier: the two-pace world, a massive record's limiting pace c_m below c (Eq. (15)); light's quadratic dispersion with the cubic pattern (Eq. (16)); and a moving clock's tick from the lattice band, Eq. (18), whose departure from the continuum's $1/\gamma$ (γ the Lorentz factor) is the lattice's own signature, read in the bound clock's row [4, B].

4 From click to click

Everything compared with nature is a click, a count of clicks or a ratio of intervals between clicks; nothing inside the board is compared. This section is the chain from a birth to the counts, and the two theorems of the Outside.

The steps between two clicks. Every comparison of this paper walks one chain of six steps.

- (1) The birth: $\mathbf{E}^\top$ at an emitter's click, the born record's two levels written once on the emitter's cells; its residue u and wheel W read from the law at the birth cell (Definition 2; DECLARATION for the born profile, the law for the rest).
 - (2) The law: the one operator, at most one Link per interval, the family's pair at every Node, the coupling and the label matrix at the tools' cells (Eq. (1)).
 - (3) The click: the record's one-way inward flux into the detectors named on its ladder crosses its threshold, and the record ends whole, everywhere, at once (Definition 1; DETECTOR).
 - (4) The output: each click's detector and interval, as the experiment's output file gives them (record 1887 [2]; DETECTOR).
 - (5) A count of clicks per detector, or a ratio of intervals between clicks (COMPUTATION on clicks; nothing of the board enters).
 - (6) The identity the algebra states for that count or ratio, the form of nature compared with.
- Step (6) is the only place where a formula meets a measurement. Anything read from the board itself is a GameBoard diagnostic and enters no comparison.

Locality and discreteness Outside, two theorems. Two properties of the game above the board follow and are not assumed.

Theorem 11 (Locality Outside). *Every step of the law but the click reads a Node's own record and its six neighbours, so a change at one Node reaches Manhattan distance m at interval m and no further [3, 9.20 (A) 5]. The click is the one non-local act: the record ends at once, everywhere. The information a click yields Outside is its detector and its interval; which detector clicks is decided by the record's own residue and its own flux, and by Theorem 7 no count at one detector depends on what is done at another. So no information passes Outside faster than one Link per interval.*

Theorem 12 (Discreteness Outside). *The counts of one detector form a free abelian monoid and their ratios lie in $\mathbb{Q}$: every Outside quantity is an integer count or a ratio of counts. A click names a detector, a cube of side at least 3, never a Node inside it; two places read are two detectors, separate names. A velocity read across Δx Links and Δn intervals is the rational $\Delta x/\Delta n$; and c Outside is a bound, at most one Link per interval on every line, with $1/\sqrt{3}$ the band's pace as the wave number goes to 0 (Eq. (13)). Nothing continuous enters above the board.*

Neither is an added assumption: the first is the law's locality with the click's decision (record 1888 [2]), the second the click's definition with the detector's cube (record 1899 [2]). What is more than these is not a theorem: that the apparatus is itself inside the law, emitters and detectors regions of the one element, is the model's construction, and the paper claims only what the clicks read.

Lorentz Outside. A boost, which mixes a step in space with a step in time, is not among the 48, and no linear bijection that preserves the polyhedral cone $|\mathbf{x}|_1 \leq t$ is a boost. Lorentz's form is the symmetry of the clicks Outside, assumed of the world and arrived at from the two assumptions of the click theorem [8]; the law itself derives a moving clock's slowing from its own dispersion, Eq. (18), and nothing more of the Lorentz group.

5 From the algebra, the physics

What follows from the algebra is stated first in one paragraph, each item with its word, and then subsection by subsection.

What is proved. Exact on the GameBoard, no experiment needed [3, 9.19, 9.20 (A), 9.25]: the group of the six Ports and, within it, the group of order 24 (Theorem 1); the equivariance of the law under the 48 and the translations, a world breaking the symmetry only through its declared data; the content, the number of quanta, conserved between clicks; per family the form I and for two coupled families J , conserved exactly before the remainders, with the exact remainder identity; the local flux identity (Proposition 3); the step's bijection, the click the one deletion (Theorem 8); locality, one Link per interval; the stability condition, every mode a rotation exactly when the largest eigenvalue is below 2; one quantum, one click (Theorem 5); Born's rule for a detector to the grain $1/W$ (Theorem 6); no signalling through the click (Theorem 7); a product state gives the CHSH sum at most 2; the CHSH sum $14/5$ on the integer axes of the Bell experiment (Section 5.4); Malus's share $a^2/(a^2 + b^2)$ on an integer axis (a, b) ; the band, the cone and light's dispersion (Section 3); no self-bound body, and a moving packet's tick from its dispersion (Theorem 10, Eq. (18)). Recovered as the wave number goes to 0: $c = 1/\sqrt{3}$, isotropic at second order by the 48; Klein–Gordon's dispersion; the continuum's $1/\gamma$. Computed and not proved: the equidistribution of the law's remainders. Carried and not proved: the push. Reached under a named assumption: Lorentz's form of the clicks Outside [8].

5.1 Geometry and symmetries: the directions and $c = 1/\sqrt{3}$

The symmetries. The law is covariant under the translations of the torus and of the interval and under the 48 signed permutations of the axes: for every g in the 48 and every translation, $\mathbf{S}(g \cdot w, g \cdot \mathbf{v}) = g \cdot \mathbf{S}(w, \mathbf{v})$, w the experiment's declared data; the click commutes with g in the same sense, the detectors' cells moving with the world, provided the order within an interval is data of the world [3, 9.19 (6), 9.25 (4)]. A world breaks the symmetry only through its regions of material. The 48 are the GameBoard's stand-in for the rotation group; a boost is not among them. The six neighbours of a Node are the vertices of the octahedron $|x| + |y| + |z| \leq 1$, the causal front of one interval; its inscribed sphere, of radius $1/\sqrt{3}$, touches the eight faces on the cube's diagonals, and $1/\sqrt{3}$ is light's band pace as the wave number goes to 0 (Eq. (13)).

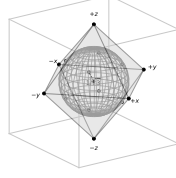


Fig. 2 The Nodes one interval from a Node: the six neighbours are the vertices of the octahedron $|x| + |y| + |z| \leq 1$, the causal front of one interval; its inscribed sphere, of radius $1/\sqrt{3}$, touches the eight faces on the cube diagonals, and $1/\sqrt{3}$ is light's band pace as the wave number goes to 0 (Eq. (13)); the faint cube shares the 48 signed permutations of the axes. Drawn by `octahedron.py` from the definitions; no run.

5.2 Conservation, the flux and momentum

The books. Between clicks nothing but a birth changes the summands, so the content is exact; a click deletes one quantum and a birth writes one, and at an emitter's click the photon and the next excited record are written as the excited record ends, so quanta are conserved at every click [3, 9.17 (4), 9.20 (A) 3]. On an open axis what leaves clicks at the face receiver and is counted there; every record clicks.

The continuity equation, exact. Summed over a region, the local flux identity (Eq. (9)) loses every Link inside the region by the antisymmetry $G_{ij} = -G_{ji}$: the change of the region's share of the form in one interval is the flux through its Ports. This is the lattice form of the continuity equation, exact in the real reading, with the remainders' exact identity beside it (DERIVED from Proposition 3).

Momentum. On a homogeneous board every character is a mode of the operator, so its wave vector is exact (PROVED [3, 1.6]); at a body the difference goes into the body by the push, CARRIED (Section 3).

5.3 Measurement and the quadratic form

The reading is quadratic by construction. The click reads the one-way inward flux, a bilinear form of the record's own two levels at the two ends of a Link, against the record's conserved form I , a quadratic form (Eqs. (4) and (9)); Born's rule, a detector's share of that flux, is a theorem (Theorem 6) and not an axiom. No principle below the square is offered: the square is the form the law conserves. A static level has $I = 0$ and clicks nowhere.

Interference, by the formulas alone. The law is linear, so two paths of one record add or cancel at a Node; a detector counts the record's inward flux, so where the two paths cancel no flux enters and no click comes. At the exit ports of a balanced Mach-Zehnder interferometer with two large detectors, one region each, the dark port's share of the inward flux is small and the bright port's large, against a half-and-half split with one arm blocked: on the placement of the lab tools' part B the shares COMPUTED by the generator's float run are 0.969 bright, 0.0062 dark (the splitters' imbalance over the packet's band and the bright beam's diffracted wing, a part that depends on the grain) and 0.025 at the faces, the spill (record 1891 [2]; [4, B]). On a screen of many detectors the counts are binomial about each detector's share of the record's total inward flux (Theorem 6); the two slits' visibility is re-derived blind on the grown board (Section 8).

Malus's share. A record with the label state $H = (1, 0)$ arriving at a polariser with the integer axis (a, b) leaves with the label rows $\mathbf{M}_a H = (a, -b)$: the component a on the detector along the axis and $-b$ on the other. The flux of each component is its form, quadratic in its amplitude, and by Eq. (8) the two sum to $(a^2 + b^2)$ times the arriving form, so by Theorem 6 the share along the axis is

$$P_+ = \frac{a^2}{a^2 + b^2}, \quad (19)$$

Malus's $\cos^2$ of the axis's angle, exact in integers [3, 9.16 (4)]: at the axes $(1, 1)$, $(5, 1)$, $(15, 8)$ and $(3, 2)$ the shares $1/2$, $25/26$, $225/289$ and $9/13$, at 45° , 11.31° , 28.07° and 33.69° degrees. There is no table of angles.

de Broglie and the rest energy. A matter record's wave number is the band's $\mathbf{k}$ at its frequency (Eq. (13)), the de Broglie fringes' spacing its consequence; the rest energy is $h\omega_0$, the pitch (Section 3).

5.4 Bell and CHSH: from weights to counts

The source. The pair is born by the crystal: the arriving record's click at the crystal's cells, then $\mathbf{E}^\top$, the birth of one record of rank 2 in the state $HV + VH$, its two arms going to two polarisers with their detectors (Definition 2; record 1879 [2]). One residue u per pair chooses the outcome for both arms at once on the joint ladder of the four outcomes, the one non-local act.

The weights, exactly. With Alice's axes $(1, 0)$ and $(1, 1)$ and Bob's $(1, 2)$ and $(3, 1)$, the joint weights of the outcomes $(++, +-, -+, --)$ are [3, 9.16 (4)]:

Alice's axis, Bob's axis	the weights	the correlation E
$(1, 0), (1, 2)$	4, 1, 1, 4 of 10	$3/5$
$(1, 0), (3, 1)$	1, 9, 9, 1 of 20	$-4/5$
$(1, 1), (1, 2)$	9, 1, 1, 9 of 20	$4/5$
$(1, 1), (3, 1)$	16, 4, 4, 16 of 40	$3/5$

The first row, worked out: the state $HV + VH$ has $w_{01} = w_{10} = 1$ and $w_{00} = w_{11} = 0$; Alice's axis $(1, 0)$ gives $\mathbf{M}_A = \mathbf{1}$ and Bob's $(1, 2)$ gives $\mathbf{M}_B = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}$, so $J(o_A, o_B) = \mathbf{M}_A[o_A][0] \mathbf{M}_B[o_B][1] + \mathbf{M}_A[o_A][1] \mathbf{M}_B[o_B][0]$ is 2, 1, 1, -2 for $(++, +-, -+, --)$, and $R = J^2$ is 4, 1, 1, 4 of 10. The correlation of one setting is

$$E = \frac{R(++) + R(--) - R(+-) - R(-+)}{\sum R},$$

$(4 + 4 - 1 - 1)/10 = 3/5$; the other three rows follow the same way (a host computation of the four weights confirms them). So the CHSH sum is

$$S = E(\mathbf{a}, \mathbf{b}) - E(\mathbf{a}, \mathbf{b}') + E(\mathbf{a}', \mathbf{b}) + E(\mathbf{a}', \mathbf{b}') = \frac{3}{5} + \frac{4}{5} + \frac{4}{5} + \frac{3}{5} = \frac{14}{5} \quad (20)$$

exactly, with Alice's $\mathbf{a} = (1, 0)$, $\mathbf{a}' = (1, 1)$ and Bob's $\mathbf{b} = (1, 2)$, $\mathbf{b}' = (3, 1)$. Tsirelson's value $2\sqrt{2}$, nature's optimum, is not a pin [3, 9.16 (4), 9.22 (6)].

Proposition 8 (a product state is local). *If the pair's weights factor, $w_{ab} = x_a y_b$, then $|S| \leq 2$ for every four axes.*

The proof is in Appendix A.

The counts as distributions. With the residue the law’s own remainder, the counts are binomial about the weights [3, 9.19 (4b)]. At $n = 100$ pair records per setting the expected counts are 40/10/10/40, 5/45/45/5, 45/5/5/45 and 40/10/10/40, and $S = 2.80$ with the standard deviation 0.141. The variance follows from the binomial counts: at one setting the estimate of E is the mean of n values ± 1 of mean E , of variance $(1 - E^2)/n$, and the four settings are independent, so

$$\text{Var}(S) = \sum_{\text{settings}} \frac{1 - E^2}{n} = \frac{16 + 9 + 9 + 16}{25n} = \frac{2.00}{n}, \quad \sqrt{2/100} = 0.141.$$

S lies above 2 by $0.8/0.141 = 5.7$ standard deviations; the verdict “above 2 at five standard deviations” needs $n \geq 79$ per setting [3, 9.19 (4b)]. The pin, written before any run: S in $[2.38, 3.22]$ (three standard deviations), the falsifier $S \leq 2$.

The marginals and the order of the outcomes. Each party’s share of $+$ is $1/2$ at every axis for the state $HV + VH$ (COMPUTED), and by Theorem 7 it does not depend on the far axis, exactly in integers; each party’s counts are binomial about it. Alice’s outcome is chosen on the outer segments of the joint ladder, whose union does not depend on Bob’s axis, so her record-by-record outcomes and click intervals do not depend on his setting (the property test’s seventh test [3, 9.25 (8)]). Whether the order of Bob’s outcomes, chosen on the inner segments, carries Alice’s setting under the law’s remainder is to be re-derived; this paper claims nothing for it.

6 The GameBoard: the engine that computes the algebra

This section states the law as the engine runs it: the one step and the two acts, what an experiment declares, the three roles, the input file and its check, the runner and the tests. Its source is the algebra document’s chapter 9 [3, 9.19 to 9.22] and the model owner’s decisions of 2026-09-25 [2].

6.1 The law and its implementation

The one step. The generic engine is one step of **S** on the one element, then the click on the summands whose ladders name a detector: the flux **E**, the rung, the deletion, and the birth **E**^T at a click that births [3, 9.19 (5)]. There is no place in it for a drive, a source term, a take, a grace window, a close, a table, a declared rate or a declared residue [3, 9.21 (3)].

Rules, parameters, initial conditions. The rules are the one operator, the click and the birth, the same for every experiment. The parameters are an experiment’s declared integers: the families’ pairs and label modules; the material map of pairs, couplings and axes per region; the emitters’ stocks and born profiles; the crystal’s branches; the occupation; every entry’s momentum **K**; and the detectors’ names and the ladders [3, 9.21 (1), (2)]. The initial conditions are not declared: the generator computes them from the operator (Eq. (11)). A composed operator with a mode at or above $2 \cos \omega = 2$ is refused at load, so no experiment carries a growing mode (the stability condition of Section 2).

The three roles. Three roles on the board, and no fourth [3, 9.22 (5)]; records 1879 and 1881 [2]. The *board generator* is the only writer, and it writes at interval 0 only: from the experiment’s short list it composes the operator and computes the initial element, the bound modes’ integer profiles and the born pairs and profiles. The *law* acts at every Node from its own values and its six

neighbours', at most one Link per interval; the birth is the law's own write, the click's other side. The *click reader* is the only reader. There is no world writer: an experiment is a short file of what is on the board, and one generic input-file generator writes one input file per experiment, naming the law it was made under.

The input file and the runner. The initial state is stored once in the experiment's input file and checked exactly at load, in integers, as Section 2 states (Eq. (12); record 1886 [2]). One command takes the input files and runs each experiment in its own process, in parallel on separate cores, the runs sharing nothing; each run checks its input (LAWFUL, or REFUSED with the reason), runs the law and writes one output file: the clicks, each its detector and interval, and the registered comparison with its blind pin, its band and the verdict; nothing else is compared (record 1887 [2]).

The three tests, rule by rule. A rule enters the law only if it is generic (one primitive with declared integers, no family name), vector (a verb on the state, no root, no float) and local (its own record and the six neighbours, nothing kept at a Node). Read on the one operator [3, 9.20 (A)]: equivariance under the 48 and the translations, PROVED for the rule, the coupling and the reading; conservation of the content DERIVED, of I and J PROVED, of momentum on a homogeneous board PROVED and at a body CARRIED; reversibility except the click PROVED; locality PROVED, the click's deletion the law's one non-local act; only the click reads, DERIVED.

The property test. The algebra's verification comes first, then the same properties computed on planted boards that are none of the experiments, then the engine's test built from them [3, 9.20 (B), (C)]: equivariance under each of the 48 bit for bit, translation, conservation between clicks with the remainder identity, reversibility under the exact inverse with no detector and the deleted record with one, locality, and no signalling (two Bell worlds differing only in Bob's axis give at Alice's detectors identical counts and click intervals, record by record). The tools' checks are these property tests on planted boards, never an exploratory reading of the GameBoard.

The cost. Per interval, per alive record, per family, every Node of the record's rows costs six reads, three multiplications and one division with its remainder; the cost is linear in the Nodes, with no global sum, and the storage is three integers per Node per row. The generator's iterations are each the work of one board step. The click reads the flux through its detectors' Ports; its deletion drops the record's rows, a host memory operation and never a computation across the board [3, 9.21 (8)]. The ladder's integers are bounded at load, the one thing that would otherwise fail silently.

7 How the algebra was reached: from the GameBoard to the group

The order of this paper is not the order of the finding. The object of Section 2 was reached from the GameBoard: from one Node and its six neighbours, one message per Link per interval, through the simulator's runs, in dated steps; what did not converge to the one operator was retired. This section is the road, marked as the road.

The one choice, and the seven Nodes. The one choice of the model is the six Ports of a Node, $P = \{\pm x, \pm y, \pm z\}$ with the opposite involution: a Node and its six neighbours, the seven Nodes of the causal front, at most one Link per interval. The translation group of the torus is the board; the point group is then forced and is already the largest [3, chapter 7].

What the choice forces: the 48, and the 24 within them. Every map that preserves the six Ports as a set and the lattice's Links is one of the 48 signed permutations of the axes, and no three-dimensional lattice has a point group of order above 48. The only $\mathbb{Z}$ -linear bijections that

preserve the cone of one Link per interval are these 48; no boost preserves the polyhedral cone $|\mathbf{x}|_1 \leq t$, whose linear automorphisms are finite, so Lorentz's form is reached Outside from the clicks and not as a symmetry of the lattice. The determinant splits the 48 into the 24 rotations, the group of order 24, and the 24 reflections; the hand, needed to carry polarisation, is the pseudoscalar that tells them apart.

From the operations to the group, the road. The operations came first, as acts on the board: in the engine before 2026-09-23 a message flew one Link per interval along its direction, messages met and merged at a Node, a body stepped by an accumulator against a wall, a phase turned on a declared circle, and a detector clicked. The 48 first appeared in the code as the invariance of one of those acts and were recognised after as the symmetry of all of them. Each act was then written as an operation on a state vector, and the operations converged, on 2026-09-25, to one linear operator on one element with its click and its birth; the stepping body, the merge, the phase circle and its tables are the road and not the law [6].

The transition, dated. The steps, each a thing of physics becoming an object of the law, with their dates [6, 2]: 2026-09-17 the shared resource deleted; 09-18 a Node its six Ports; 09-19 the quantum an integer record and the click the one border; 09-20 the readings by kind; 09-21 the three tests of every rule and the lattice the translation group; 09-22 the detector's clock and the click theorem; 09-23 the massive record kind, one pair per family, and the local detector law; 09-24 the tools as operations of the group and the reading as the one-way flux; 09-25 the whole board one element and the law one operator (record 1875), no table and no formula on the board (1878), the crystal and the three roles (1879), the input file and its load check (1881, 1886), the moving body already moving and the moving name (1884, 1889), the click the only non-local act (1888), and a detector's sensitivity its cube (1899).

The platform: how a formula is arrived at. Every formula of this paper is arrived at by two roads that meet at the detector. The algebraic road: the one operator on the experiment's entries, the conversion by an emitter's birth and a detector's click, and the Outside form, a count or a ratio of intervals, derived before the run. The simulator's road runs the same steps: the experiment's list, the generator's input file, the engine's run, the output file's clicks. The two meet in the pin, the algebra's number written blind before the run, and the clicks, read after it.

8 The comparison with nature, and the failures by cause

Everything compared here is a click, a count of clicks or a ratio of intervals between clicks (Section 4); each failure will be named with its cause.

8.1 The simulator's checks and the comparison with nature

The method. An experiment is a list of entries on the board (Section 2, "The experiment as data"): the board and its faces, the families, the tools placed as cubes with their material integers, the occupation, every entry's momentum and the detectors' names; the generator writes its input file, the engine checks it and runs it, and its result is its clicks. The pin, the number the algebra gives, is written blind before the run, and no pin moves after a reading. Every experiment passes three checks in order: approved in the algebra, its pin following from the one operator before any run [3, 9.22 (8)]; the operator and its tools approved by the property test on planted boards and the load check, never by an experiment's run; and the pinned run, its clicks read by the click reader and compared with the pin within its band. An experiment is (K), a known-formula row, or (P), a

prediction row (Section 3); a control checks another row, and a row read by a GameBoard probe is a diagnostic, not compared with nature.

What lies between the tools. Every experiment states what its real apparatus has between its tools (air, a vacuum chamber, a fibre, a holder), that the board neglects it, the space between the tools being the vacuum of the one operator, and why, as a number against the pin’s band; where the neglect is not below the band, the medium enters as a material region with its own pair, never as a silent omission (the model owner’s decision, record 1891 [2]). Each experiment’s statement, with its number against its band, is written in the lab tools’ part B [4, B]: for every row but one the neglect is below the band; the muon’s slowing in air is not, and that row declares the idealisation of a fixed momentum.

The twenty-four experiments. The paper has twenty-four experiments (the model owner’s decision, record 1915 [2]): eighteen measured, run on the board and read in clicks against their blind pins, and six algebraic, whose result is the algebra’s (Section 8.2). Table 1 lists the eighteen measured experiments (records 1883, 1891 and 1911 [2]): the fifteen, the two-qubit computer, Mach–Zehnder and Sorkin’s three openings. The in-motion halves of the measured experiments whose boards are heavy keep their closed forms inside their own rows: the closed form stands if the run is too heavy, and a run must agree with it. Each row gives what is on the board, the pin written blind with its source, and the row’s status. Ten rows carry a pin written blind (rows 3, 4, 5, 7, 9, 10, 13, 14, 16 and 17); seven have their pin still owed, to a re-derivation on the grown board (row 1), on the flux form (rows 6, 8 and 11) or on the bound medium (rows 12 and 15), or to the mathematician’s first derivation (row 18); row 2 is a diagnostic with a closed form and no click pin. No row is run on the new engine yet: every reading of the engine before 2026-09-25 has left the body, and every row is held until its pin is re-derived blind and its run is read.

Table 1 The eighteen measured experiments on the new engine. On the board: the tools as regions of the one operator. The pin: the algebra’s number written blind before any run, with its source in the algebra document [3, 9.22 (8)] or the lab tools’ part B [4, B]; “owed” marks a pin whose re-derivation on the flux form or on the regenerated board is still to be written. Kind: (K) a known formula compared with, (P) a prediction, ctrl. a control, diag. a diagnostic: a row read by a GameBoard probe, which under the clicks-only rule [3, 9.21 (4)] is not compared with nature until its reading has a click form. Status: every row is held, re-derived blind on the new engine, its comparison with nature to follow its run.

#	Experiment	On the board	The pin, written blind (source)	Kind	Status
1	The two slits	an emitter body with a stock of 1024; a wall of gap cubes with two openings; a screen of 67 detector cubes of side 3; the face receivers; the board grown for its margin	the visibility 0.96 ± 0.02 , owed on the grown board ([3, 9.22 (8)], row 3; [4, B])	K	held: re-derived blind on the new engine
2	The pace fans	two line emitters, one along an axis and one along a diagonal	$k(\text{axis}) - k(\text{diagonal}) = k^2/48$ in closed form, a GAMEBOARD diagnostic with no click pin ([3, 9.22 (8)], row 4)	diag.	held: re-derived blind on the new engine
3	The muon’s form	at rest a well of side 14; in motion a packet of the matter family with K for $v = 1/3$ and a moving name	at rest the mode’s period 42.33; in motion $f/f_0 = 0.8146$ from the dispersion, band 0.3 percent ([3, 9.22 (8)], row 5; [4, B])	K	held: re-derived blind on the new engine
4	The redshift	at rest a well with the coupling and a stock; in motion the emitter a packet with a moving name; a receiver	$1 + z = 1.935$ at $v = 1/3$, band 0.3 percent ([3, 9.22 (8)], row 6; [4, B])	K	held: re-derived blind on the new engine
5	The round trip	a material mirror at rest; the emitter and its receiver one moving name on a packet receding from it	$(1 + \beta)/(1 - \beta) = 3.7373$, β the speed over light’s group velocity, band 0.3 percent ([3, 9.22 (8)], row 7; [4, B])	K	held: re-derived blind on the new engine

#	Experiment	On the board	The pin, written blind (source)	Kind	Status
6	Sagnac	two wells with couplings and stocks and two detectors; co-moving, two packets with moving names	at rest the first rung 108 ± 2 , owed on the flux form; in motion the first rungs scaled by $c_l/(c_l \mp v)$, c_l light's group velocity, owed ([3, 9.22 (8)], row 8; [4, B])	K	held: re-derived blind on the new engine
7	de Broglie's fringes	a matter emitter body with a stock of 2048; a wall of gap cubes with its openings; a screen of 41 detector cubes of side 3; the face receivers	the side lobe's centroid at $64 + 27.4$, band one Node ([3, 9.22 (8)], row 9; [4, B])	K	held: re-derived blind on the new engine
8	The moving mass	a matter emitter body with a stock; a receiver; the face receivers	the first rung 169 ± 2 , owed on the flux form ([3, 9.22 (8)], row 10; [4, B])	K	held: re-derived blind on the new engine
9	The bound clock's second term	at rest a well of side 20; in motion a packet with K for $v = 1/3$ and a moving name	at rest the mode's clock; in motion $f/f_0 = 0.8146$ from the dispersion, band 0.3 percent ([3, 9.22 (8)], row 11; [4, B])	P	held: re-derived blind on the new engine
10	The deep well in motion, and its control	at rest a well of side 40; in motion a packet as in row 9	at rest the mode's clock; in motion $f/f_0 = 0.8146$, the control ([3, 9.22 (8)], row 12; [4, B])	ctrl.	held: re-derived blind on the new engine
11	The light clock	a well with the coupling and a stock; a material mirror; a receiver; the face receivers	at rest 214 ± 1 , the closed form $2D/v_g$ (D the arm's length in Links, v_g light's group velocity) plus the mirror's delay with the rung's offset, owed on the flux form; in motion a closed form only, never run ([3, 9.22 (8)], row 13; [4, B])	K	held: re-derived blind on the new engine
12	The receding index at $k = 3$	a light emitter body; a medium body of the holder family, bound and at rest, with the coupling; the emitter and the probe a moving name	the closed form $n^2 = 1 + Gg/(\omega_0^2 - \omega^2)$ with the phase drift, owed on the bound medium; the probe a GAMEBOARD reading ([3, 9.22 (8)], row 14; [4, B])	diag.	held: re-derived blind on the new engine
13	Bell	an emitter body; the crystal; two polariser bodies with integer axes; four detectors	S in [2.38, 3.22] at $n = 100$ pair records per setting, the expected counts $40/10/10/40$, $5/45/45/5$, $45/5/5/45$, $40/10/10/40$; the falsifier $S \leq 2$ ([3, 9.19 (4b), 9.22 (8)], row 1)	K	held: re-derived blind on the new engine
14	Malus	an emitter body with a stock of 256; a polariser body with an integer axis; its detector	at 256 records on the + detector: 128 ± 24 at (1, 1), 246 ± 9 at (5, 1), 199 ± 20 at (15, 8), 177 ± 22 at (3, 2) ([3, 9.19 (4b), 9.22 (8)], row 2)	K	held: re-derived blind on the new engine
15	The receding index at $k = 4$	as row 12, at $k = 4$	as row 12 ([3, 9.22 (8)], row 15; [4, B])	diag.	held: re-derived blind on the new engine
16	The two-qubit computer	an emitter body; the crystal with the prepared state's branches; two readout polarisers with their detectors	Grover's search 0/0/0/100 at $n = 100$; Deutsch-Jozsa 100/0 and 0/100 per oracle; the band 0 ([3, 9.22 (8)], row 16; [4, B])	K	held: re-derived blind on the new engine
17	Mach-Zehnder	an emitter body; two splitters and two mirrors, material regions; two large detectors, one connected region at each exit port	at 100 records: the bright port 97 ± 5 (share 0.969), the dark port 0 or 1 (share 0.0062; two at the 12 percent level, three or more a defect), the faces 3 (0 to 7, share 0.025, the spill); the control with one arm blocked: bright 47 ± 15 , dark 39 ± 15 , faces 14 ± 10 ([4, B], rows 17 and its control)	K	held: re-derived blind on the new engine
18	Sorkin's three openings	seven worlds on the two slits' placement: the three openings, the three pairs and each opening alone, with the screen's detectors	owed: the three-opening sum in clicks with its statistical band, written blind by the mathematician before any run (record 1911 [2])	K	held: re-derived blind on the new engine

The two-qubit computer and Mach-Zehnder. The sixteenth experiment is a two-qubit quantum computer in its minimal form: the circuit's computation is done by the input-file generator, the gates iterated on the weights at a cost exponential in the qubits and outside the board, and written as the crystal's branches; the board carries the entanglement locally to two readouts and the clicks read it, so no new engine form is needed (records 1883 and 1890 [2]; [3, 9.23]). The

seventeenth shows interference by clicks alone: two large detectors at the exit ports of a balanced Mach–Zehnder interferometer, every tool a region of the one operator; the dark port’s computed share is 0.0062 and the faces’ spill 0.025, against a bright share of 0.969; with one arm blocked the two detectors are lit alike within their bands (record 1891 [2]; [4, B]).

The limitations, stated before the comparison. The law derives a moving clock’s slowing from the dispersion (Eq. (18)) and not the Lorentz group: the GameBoard is a medium with a rest frame, and Lorentz’s symmetry belongs to the clicks Outside. The model has clocks and no rods: a moving rigid arm is not defined, so no Michelson–Morley or Kennedy–Thorndike row is claimed and the longitudinal light clock in motion is not claimed (record 1895 [2]). A region of material does not move, so a moving tool is a packet with a moving name. A declared input is not a prediction. Every comparison is a count or a ratio of intervals, so no unit enters; the paper fixes no Link’s physical length.

The rows that left the list. Under the new engine the following are not in the list, their worlds cancelled [9]: GHZ, the uncertainty relation, the equivalence principle, the bending of light and the delay, the pair-field rows, the quarks and the dark sector; the single opening’s spread and the light clock in motion along its arm. The atom’s lines, the bound on the length of one interval and gravity’s forms are not measured: they are algebraic experiments (Section 8.2).

The prediction rows. The (P) rows are the algebra’s own numbers with no known formula behind them: the bound clock’s tick in motion from the lattice band (row 9), with the two paces and light’s dispersion pattern (Section 3) as predictions of form.

8.2 The six algebraic experiments

Six experiments are algebraic: their result is the algebra’s, computed without a run, and each is stated with how it is computed, the model’s number beside nature’s and whether they agree. Every number of this section is a COMPUTATION, never a measurement, and none is compared as a click (the model owner’s decision, record 1915 [2]). The mathematician writes each row’s computation; until he does, its numbers are owed and none is written here.

Table 2 The six algebraic experiments, each a COMPUTATION of the algebra and never a measurement.

#	Experiment	How it is computed	The model’s number; nature’s	Agree?
A1	The atom’s lines	the bound modes’ own frequencies of a well, where nature’s lines are at the differences of levels	owed (the mathematician)	a stated disagreement
A2	Light’s dispersion by direction and wavelength	the coefficient of Eq. (16), $(3 \sum_i n_i^4 - 1)/24$, against the bounds from astrophysical bursts	owed (the mathematician)	owed
A3	The bound on the length of one interval	from the two paces of Eq. (15) and light’s dispersion, against nature’s bounds	owed (the mathematician)	owed
A4	The bound clock’s second term	the closed form of a bound clock’s departure from the continuum’s dilation (the measured row 9 reads the tick by clicks)	owed (the mathematician)	owed
A5	Born’s exponent and Tsirelson’s limit	the algebra’s identities: the exponent 2 of the reading’s form (Theorem 6) and the CHSH sum’s limit	owed (the mathematician)	owed
A6	Gravity’s forms	Newton’s inverse square and Poisson’s equation under a shell average, as limits of the algebra and not as a run	owed (the mathematician)	owed

8.3 Discussion and conclusions

Positioning. The model is a local deterministic lattice dynamics with one non-local act, the click; its correlations violate the CHSH bound [10, 11] through that act, and its counts do not signal [12] (Theorem 7). Its transport is an exact integer lattice dynamics, as a lattice gas’s is [13, 14], and its light disperses at k^2 with the cubic pattern where the lattice automata of [15, 16, 17] disperse by their own forms. The residue read from the law does the work of quantum equilibrium [18, 19]; the model is ψ -ontic (the wave function is part of the state) [20, 21]; its CHSH sum on integer axes lies between the local bound 2 and Tsirelson’s $2\sqrt{2}$ [22, 23]. The earlier testbed [1] is history.

Conclusion. One local integer operator on the cubic lattice, required to commute with the group of order 48, with one non-local act, the click, and one write, the birth, is read only through detectors’ clicks. Exact on the lattice are the conserved forms, the step’s bijection, the flux identity, one click per quantum, Born’s rule for a detector and no signalling; the CHSH sum on integer axes is $14/5$; a moving clock slows by its own dispersion. Of twenty-four experiments, eighteen are measured, ten of them with a pin written blind, seven with it still owed and one a diagnostic (Table 1), and six are algebraic (Table 2); none is run yet on the new engine, and the runs decide, row by row, what the law matches and where it fails.

Declarations

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Ethics approval. Not applicable.

Consent to participate and consent for publication. Not applicable: no human participants and no animals.

Data availability. Every experiment’s short file and input file (with the law it was made under and its hash), every output file with its clicks and its verdict, the generator and the derivation records cited in this paper are archived at Zenodo [24] (concept DOI 10.5281/zenodo.22738746); the commit of the archived version is named in Appendix C, and every number of the paper names its section of the algebra document or of the lab tools’ specification.

Materials availability. Not applicable: no physical materials.

Code availability. The simulator, the input-file generator and the figures’ scripts are in the same archive [24], under its open licence; the figures are drawn by those scripts from the definitions, none by a generative tool.

Author contributions. Sole author: the guidelines of the project, the model, its rules, the worlds and the experiments, every reading and its verdict, and the decisions recorded in the archived log; the part of the large language model used as a tool is stated in the next declaration.

Use of artificial intelligence. The author used a large language model (Claude, Anthropic, through Claude Code, 2026) extensively as a tool, under the author’s direction and instructions, for the software of the simulator and its checks, the computations, the mathematical derivations and their verification, the physics of the design pages, and the drafting and editing of the text; the model is not an author and meets no criterion of authorship. Every guideline, postulate, hypothesis,

rule, world and decision of the project is the author's, recorded in the archived log; the author reviewed and edited every statement of this paper and takes full responsibility for its content. The author's statement: I laid down every postulate and hypothesis of the model myself and brought in nothing external; the known forms that this paper cites are the things compared with, not inputs to the model.

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A The proofs in full

The proofs of the statements of the body, in the order they stand there, each from the algebra document's section cited in it.

Proof of Theorem 1. A map that keeps opposite Ports opposite is fixed by the images of $+x$, $+y$, $+z$, which must be three Ports no two of them equal or opposite: $6 \cdot 4 \cdot 2 = 48$ choices, each a signed permutation matrix, and every signed permutation matrix keeps opposite Ports opposite. The determinant of a product is the product of the determinants, so $\det$ is a homomorphism, onto since $-\mathbf{1}$ has determinant -1 ; its kernel has index 2, order 24. A rotation permutes the four lines through opposite vertices of the cube, and only the identity fixes all four, so the kernel embeds in S_4 , of order 24: the two are isomorphic. $\square$

Proof of Theorem 2. Write $\mathbf{x} = \mathbf{a}_{\text{now}}$, $\mathbf{y} = \mathbf{a}_{\text{next}}$ and $\mathbf{y}' = \mathbf{M}\mathbf{y} - \mathbf{x}$ the level one interval later, with no remainder. Then

$$I(\mathbf{y}', \mathbf{y}) - I(\mathbf{y}, \mathbf{x}) = \mathbf{y}' \cdot \mathbf{D}\mathbf{y}' - \mathbf{x} \cdot \mathbf{D}\mathbf{x} - \mathbf{y}' \cdot \mathbf{D}\mathbf{M}\mathbf{y} + \mathbf{y} \cdot \mathbf{D}\mathbf{M}\mathbf{x}.$$

Since $\mathbf{D}\mathbf{M}\mathbf{y} = \mathbf{D}\mathbf{y}' + \mathbf{D}\mathbf{x}$, the third term is $-\mathbf{y}' \cdot \mathbf{D}\mathbf{y}' - \mathbf{y}' \cdot \mathbf{D}\mathbf{x}$. Since $\mathbf{D}\mathbf{M} = \mathbf{A}/3$ is symmetric, the fourth term is $(\mathbf{D}\mathbf{M}\mathbf{y}) \cdot \mathbf{x} = \mathbf{y}' \cdot \mathbf{D}\mathbf{x} + \mathbf{x} \cdot \mathbf{D}\mathbf{x}$. The four terms sum to 0. Nothing but the symmetry of $\mathbf{D}\mathbf{M}$ is used, so the identity holds on every extent and with every face. With the remainder, $\mathbf{y}' = \mathbf{M}\mathbf{y} - \mathbf{x} + \mathbf{e}$, so $\mathbf{D}\mathbf{M}\mathbf{y} = \mathbf{D}\mathbf{y}' + \mathbf{D}\mathbf{x} - \mathbf{D}\mathbf{e}$: the third term gains $+\mathbf{y}' \cdot \mathbf{D}\mathbf{e}$ and the fourth gains $-\mathbf{e} \cdot \mathbf{D}\mathbf{x}$, and the change is $\mathbf{e} \cdot \mathbf{D}(\mathbf{y}' - \mathbf{x})$. In the interval that $\mathbf{y}'$ ends, $\mathbf{y}'$ is $\mathbf{a}_{\text{next}}$ and $\mathbf{x}$ is $\mathbf{a}_{\text{before}}$, which is Eq. (5) [3, 8.2]. $\square$

Proof of Proposition 1. Let $\mathbf{N} = \mathbf{D}^{-1/2}(\mathbf{A}/3)\mathbf{D}^{-1/2}$, symmetric. In the variables $(\mathbf{D}^{1/2}\mathbf{a}_{\text{next}}, \mathbf{D}^{1/2}\mathbf{a}_{\text{now}})$ the form I is the quadratic form of the block matrix $\begin{pmatrix} \mathbf{1} & -\mathbf{N}/2 \\ -\mathbf{N}/2 & \mathbf{1} \end{pmatrix}$, $\mathbf{1}$ the identity; on an eigenvector of $\mathbf{N}$ with the eigenvalue λ it is $\begin{pmatrix} 1 & -\lambda/2 \\ -\lambda/2 & 1 \end{pmatrix}$, with the eigenvalues $1 \pm \lambda/2$, positive exactly when $|\lambda| < 2$. By Gershgorin every eigenvalue of $\mathbf{A}$ has $|\lambda| \leq 6$ (each row at most six entries, each 1), so $|\lambda(\mathbf{N})| \leq 6/(3 \min_i D_i) = 2 \max_i (\text{num}_i / \text{den}_i)$, below 2 when every $\text{den}_i > \text{num}_i$ and equal to 2 at light's pair, whose null directions are the zero mode and the checkerboard [3, 8.2]. $\square$

Proof of Proposition 2. The derivative of $L(t-1) + L(t)$ with respect to $\mathbf{a}_{m,t}$, set to zero, is $-\mathbf{D}\mathbf{a}_{m,t-1} + g\mathbf{D}\mathbf{P}_R(\mathbf{a}_{l,t} - \mathbf{a}_{l,t-1}) - \mathbf{D}\mathbf{a}_{m,t+1} + \frac{1}{3}\mathbf{A}\mathbf{a}_{m,t} = 0$, which is the massive step with light's backward difference. With respect to $\mathbf{a}_{l,t}$ it is $-\alpha\mathbf{a}_{l,t-1} + g\mathbf{P}_R\mathbf{D}\mathbf{a}_{m,t} - g\mathbf{P}_R\mathbf{D}\mathbf{a}_{m,t+1} - \alpha\mathbf{a}_{l,t+1} + \alpha\frac{1}{3}\mathbf{A}\mathbf{a}_{l,t} = 0$, that is $\mathbf{a}_{l,t+1} + \mathbf{a}_{l,t-1} = \frac{1}{3}\mathbf{A}\mathbf{a}_{l,t} - (gD_{\text{in}}/\alpha)\mathbf{P}_R(\mathbf{a}_{m,t+1} - \mathbf{a}_{m,t})$, light's step with $gD_{\text{in}}/\alpha = G$ [3, 8.5]. $\square$

Proof of Theorem 4. The massive step is Eq. (3) with $\mathbf{e}_m = g\mathbf{P}_R(\mathbf{a}_{l,\text{now}} - \mathbf{a}_{l,\text{before}})$, so by Eq. (5) $I_m(t) - I_m(t-1) = gD_{\text{in}} \sum_{i \in R} (a_{l,\text{now}} - a_{l,\text{before}})_i (a_{m,\text{next}} - a_{m,\text{before}})_i$. Light's step has $\mathbf{e}_l = -G\mathbf{P}_R(\mathbf{a}_{m,\text{next}} - \mathbf{a}_{m,\text{now}})$, so $\alpha(I_l(t) - I_l(t-1)) = -gD_{\text{in}} \sum_{i \in R} (a_{m,\text{next}} - a_{m,\text{now}})_i (a_{l,\text{next}} - a_{l,\text{before}})_i$. Write per cell $X = a_{m,\text{next}} - a_{m,\text{now}}$, $Y = a_{m,\text{now}} - a_{m,\text{before}}$, $U = a_{l,\text{next}} - a_{l,\text{now}}$ and $V = a_{l,\text{now}} - a_{l,\text{before}}$. The cross term changes by $gD_{\text{in}}(XU - YV)$, and the whole change of J per cell is

$$gD_{\text{in}}[V(X+Y) - X(U+V) + XU - YV] = 0$$

[3, 8.5]. $\square$

Proof of Proposition 3. Write $x_i = a_{\text{next},i}$, $y_i = a_{\text{now},i}$ and $z_i = a_{\text{before},i}$, so $e_i(t) = D_i(x_i^2 + y_i^2) - \frac{1}{3}x_i(\mathbf{A}\mathbf{y})_i$ and $e_i(t-1) = D_i(y_i^2 + z_i^2) - \frac{1}{3}y_i(\mathbf{A}\mathbf{z})_i$. Their difference is $D_i(x_i^2 - z_i^2) - \frac{1}{3}[x_i(\mathbf{A}\mathbf{y})_i - y_i(\mathbf{A}\mathbf{z})_i]$. The rule gives $D_i(x_i + z_i) = \frac{1}{3}(\mathbf{A}\mathbf{y})_i$, so $D_i(x_i^2 - z_i^2) = \frac{1}{3}(x_i - z_i)(\mathbf{A}\mathbf{y})_i$, and the difference is

$$\frac{1}{3}[(x_i - z_i)(\mathbf{A}\mathbf{y})_i - x_i(\mathbf{A}\mathbf{y})_i + y_i(\mathbf{A}\mathbf{z})_i] = \frac{1}{3}[y_i(\mathbf{A}\mathbf{z})_i - z_i(\mathbf{A}\mathbf{y})_i] = \sum_j \frac{1}{3}A_{ij}(y_i z_j - z_i y_j),$$

which is Eq. (9); the pairs have cancelled, and $G_{ji} = -G_{ij}$ since $\mathbf{A}$ is symmetric [3, 9.19 (3)]. $\square$

Proof of Theorem 5. Every increment $f_d(t) \geq 0$, so C is non-decreasing. Since $u \leq W - 1$, $\theta = (2u + 1)T/(2W) \leq (2W - 1)T/(2W) < T$. With the face receiver last on the ladder, a record's whole form is eventually offered: what stays on the board enters some detector, and what leaves enters the face, so $C(t) \rightarrow C(\infty) \geq T > \theta$ and θ is crossed at a first interval t^* . At t^* the record is deleted whole, so no later increment exists and θ cannot be crossed again: one click [3, 9.25 (2)]. $\square$

Proof of Theorem 6. With u uniform on $\{0, \dots, W - 1\}$ the threshold $\theta = (2u + 1)T/(2W)$ is uniform on the W midpoints of a partition of $[0, T)$ into cells of length T/W . Lay out, for each interval t and each detector d of the ladder in its declared order, the segment

$$\left[C(t-1) + \sum_{d' \text{ before } d} f_{d'}(t), C(t-1) + \sum_{d' \text{ up to } d} f_{d'}(t) \right)$$

of length $f_d(t)$. These segments are disjoint and cover $[0, C(\infty))$, and $C(\infty) \geq T$ with the face last. By Definition 1 the click falls at (t, d) exactly when θ lies in that segment, so the number of the W midpoints in it is $Wf_d(t)/T$ up to one at each end, and $P(t, d) = f_d(t)/T$ to the grain $1/W$. Summing over t , $P(d) = \sum_t f_d(t)/T = C_d(\infty)/T$. The total length of a detector's segments is $\sum_t f_d(t)$ for every order within an interval, so the distribution does not depend on that order; a single record's detector at a given u does, which is why the order is the world's declared data [3, 9.25 (3), (4)]. A detector's increment is the sum of its boundary Nodes' inward fluxes from outside, so its segment is the union of its Nodes' segments and its share is the sum of their shares [3, 9.25 (10)]. $\square$

Proof of Theorem 7. Let the pair's four weights be w_{ab} , a Alice's label and b Bob's, and let $\mathbf{M}_A$ and $\mathbf{M}_B$ be the two label matrices of the axes (a_A, b_A) and (a_B, b_B) . The joint amplitude is

$$J(o_A, o_B) = \sum_{a,b} w_{ab} \mathbf{M}_A[o_A][a] \mathbf{M}_B[o_B][b],$$

and the weight is $R = J^2$. For a fixed o_A write $(\mathbf{w}_{o_A})_b = \sum_a w_{ab} \mathbf{M}_A[o_A][a]$, her arm's label state projected on her own outcome; then $J(o_A, o_B) = (\mathbf{M}_B \mathbf{w}_{o_A})_{o_B}$ and

$$\sum_{o_B} R(o_A, o_B) = |\mathbf{M}_B \mathbf{w}_{o_A}|^2 = (a_B^2 + b_B^2) |\mathbf{w}_{o_A}|^2$$

by Eq. (8). So her share of $+$ is $(a_B^2 + b_B^2) |\mathbf{w}_+|^2 / ((a_B^2 + b_B^2)(|\mathbf{w}_+|^2 + |\mathbf{w}_-|^2))$, the scale cancelling exactly. On the joint ladder with o_A outer, her outcome $+$ is the union of the segments $[0, \sum_b R(+, b))$, whose length is fixed by her own projection whatever Bob's axis; her click's time is her own arm's flux through her own detectors. Nothing of Bob's setting, placement or distance enters either [3, 9.25 (6)]. $\square$

Proof of Theorem 8. Fix S_6 and write $d = 3 \text{ den}$. The map $(a_{\text{before}}, r) \mapsto n = \text{num } S_6 - d a_{\text{before}} + r$ from $\mathbb{Z} \times \{0, \dots, d - 1\}$ to $\mathbb{Z}$ is a bijection: every integer n has exactly one representation $n - \text{num } S_6 = -d a_{\text{before}} + r$ with $0 \leq r < d$, the Euclidean division of $\text{num } S_6 - n$ by d read with the ceiling. The map $n \mapsto (\lfloor n/d \rfloor, n \bmod d) = (a_{\text{next}}, r')$ is a bijection of $\mathbb{Z}$ onto $\mathbb{Z} \times \{0, \dots, d - 1\}$. Their composition is the step, and its inverse is the pair of inverses: from (a_{next}, r') , $n = d a_{\text{next}} + r'$, $m = \text{num } S_6 - n = d a_{\text{before}} - r$, so $a_{\text{before}} = \lceil m/d \rceil$ and $r = d a_{\text{before}} - m$. The reads S_6 are of the level now, which the inverse holds at every Node, so the inverse is computed Node by Node. For two coupled families each half-step is invertible given the other family's levels, which it leaves in place, so the pair of half-steps is inverted in the reverse order [3, 8.8]; [3, 9.20 (A) 4]. $\square$

Proof of Proposition 5. $\mathbf{M} + 2\mathbf{1}$ has the eigenvalues $\lambda_k + 2$ of $\mathbf{M}$ shifted by 2, all in $(0, 4)$ by Theorem 3, so they are positive and the lowest mode cannot compete in absolute value. Its entries are nonnegative, so by Perron–Frobenius its top eigenvalue is simple on a connected board with a nonnegative eigenvector, the bound mode. Expanding the start on the eigenvectors, the k -th component is multiplied by $((\lambda_k + 2)/(\lambda + 2))^n$ after n iterations, at most $(1 - \text{gap}/(\lambda + 2))^n$ for every k but the top [3, 9.22 (7)]. $\square$

Proof of Proposition 6. Write $\mathbf{p} = \mathbf{q} + \boldsymbol{\delta}$ with $|\delta_i| \leq 1/2$, and $a/b = \lambda + \varepsilon$ with $|\varepsilon| \leq 1/(2b)$. Using the eigen-equation,

$$b \text{num}_i(S_6 \mathbf{p})_i - 3 \text{den}_i a p_i = b \text{num}_i(S_6 \boldsymbol{\delta})_i - 3 \text{den}_i b \lambda \delta_i - 3 \text{den}_i b \varepsilon q_i - 3 \text{den}_i b \varepsilon \delta_i.$$

The four terms are bounded by $6 \cdot \frac{1}{2} b \text{num}_i = 3b \text{num}_i$, by $3 \text{den}_i \cdot 2 \cdot \frac{1}{2} b = 3b \text{den}_i$, by $3 \text{den}_i \cdot \frac{1}{2} A \leq \frac{3}{2} b \text{den}_i$, and by $\frac{3}{4} \text{den}_i \leq \frac{3}{2} b \text{den}_i$; their sum is at most $b(3 \text{num}_i + 6 \text{den}_i)$ [3, 9.22 (7)]. $\square$

Proof of Theorem 9. By the angle-addition formula, $a(\mathbf{x}, t+1) + a(\mathbf{x}, t-1) = 2 \cos \omega a(\mathbf{x}, t)$, and the six reads give $S_6 = \sum_{\pm, i} \cos(\omega t - \mathbf{k} \cdot \mathbf{x} \mp k_i) = 2(\cos k_x + \cos k_y + \cos k_z) a(\mathbf{x}, t)$. The rule $3 \text{den}(a_{\text{next}} + a_{\text{before}}) = \text{num } S_6$ holds at every Node and interval exactly when $3 \text{den} \cdot 2 \cos \omega = 2 \text{num}(\cos k_x + \cos k_y + \cos k_z)$, which is Eq. (13). Since $|\cos k_x + \cos k_y + \cos k_z| \leq 3$, $|\cos \omega| \leq \text{num}/\text{den} \leq 1$ [3, 8.1]. $\square$

Proof of Proposition 7. At $[1, 1]$, $\cos \omega = \frac{1}{3} \sum_i \cos(k n_i) = 1 - \frac{k^2}{6} + \frac{k^4}{72} \sum_i n_i^4 + O(k^6)$, using $\sum_i n_i^2 = 1$. Write $u = 1 - \cos \omega = \frac{k^2}{6} - \frac{k^4}{72} \sum_i n_i^4 + O(k^6)$. Inverting $u = \omega^2/2 - \omega^4/24 + O(\omega^6)$ gives $\omega^2 = 2u + u^2/3 + O(u^3)$, so

$$\omega^2 = \frac{k^2}{3} - \frac{k^4}{36} \sum_i n_i^4 + \frac{k^4}{108} + O(k^6) = \frac{k^2}{3} \left(1 - \frac{(3 \sum_i n_i^4 - 1) k^2}{36} \right) + O(k^6),$$

$\omega = (k/\sqrt{3})(1 - (3 \sum_i n_i^4 - 1)k^2/72 + O(k^4))$, and $d\omega/dk$ is Eq. (16) with $c = 1/\sqrt{3}$ [7]. $\square$

Proof of Theorem 10. Away from material the step commutes with every translation of the torus, so it commutes with the discrete Fourier transform's shifts and is diagonal on the characters $e^{i\mathbf{k} \cdot \mathbf{x}}$: every solution is a sum of characters, each advanced by its own clock $\omega(\mathbf{k})$ of Eq. (13). A stationary solution, one whose level returns to itself up to a phase, is then a single character or a sum at one frequency, spread over the whole board, and not localised; a localised mode needs an eigenvector of a composed operator that breaks the translations, a region of material. A body's own quanta cannot bind each other, since that would need a term of degree two or more in one family's amplitude, which the three tests exclude. A well carried by a record of the holder family would need a family's pair at a Node to be a function of the holder's amplitude there, a term of degree three, or a threshold read on a cell, and is refused [3, 9.24 (1), (3)]. $\square$

Proof of Proposition 8. Then $J(o_A, o_B) = (\mathbf{M}_A \mathbf{x})_{o_A} (\mathbf{M}_B \mathbf{y})_{o_B}$ and R factors, $R(o_A, o_B) = P_A(o_A) P_B(o_B) \cdot (\text{a positive scale})$, so $E(\mathbf{a}, \mathbf{b}) = e_A(\mathbf{a}) e_B(\mathbf{b})$ with $|e_A|, |e_B| \leq 1$. Then $S = e_A(\mathbf{a})[e_B(\mathbf{b}) - e_B(\mathbf{b}')] + e_A(\mathbf{a}') [e_B(\mathbf{b}) + e_B(\mathbf{b}')]]$, and $|S| \leq |e_B(\mathbf{b}) - e_B(\mathbf{b}')| + |e_B(\mathbf{b}) + e_B(\mathbf{b}')| = 2 \max(|e_B(\mathbf{b})|, |e_B(\mathbf{b}')|) \leq 2$ [3, 9.22 (6)]. $\square$

B Technical details

The integers. Every level, remainder, pair, coupling, axis, born pair and profile is an integer of the world; the engine carries no table and no formula, the phase tables living only in the generator and the host's readers, outside the engine (record 1878 [2]). The ladder's integers, the norm T and the running total C scaled by 3 and by the pairs' common wall, grow with the amplitude squared times a record's support and the wheel; they are arbitrary-precision, or the amplitude is bounded at load, a load check [3, 9.21 (8d)].

The symbols. Every symbol is named at its first use in the text, a scalar plain, a vector in bold lowercase, a matrix or an operator in bold uppercase: $\mathbf{v}$ the one element; $\mathbf{S}$ the one operator; $\mathbf{A}$ the read matrix; $\mathbf{D}$ the diagonal of the per-Node weights $\text{den}_i/\text{num}_i$; $\mathbf{M} = \mathbf{D}^{-1} \mathbf{A}/3$ the rule's matrix; $\mathbf{E}$ the reading and $\mathbf{E}^\top$ the birth; $\mathbf{M}_u$ the label matrix of the axis $\mathbf{u}$; $\mathbf{K}$ an entry's momentum, the wave vector of its character.

C Reproduction

The paper's law is the one operator with its click and its birth, the algebra document's sections 9.19 to 9.25 with chapter 8, on `main` at `dfeb7abc` [3], and the lab tools' specification at the same commit [4]. Each experiment is its short file of entries; its input file, written once by the input-file generator, names the law and the generator it was made under and is checked at load (LAWFUL or REFUSED); one command runs

the input files in parallel and writes one output file per experiment, its clicks and its verdict against the blind pin (records 1881, 1886 and 1887 [2]). The engine's commit and the law's identity that each input file carries are named here with the runs. The software, the computations, the derivations' checks and the text were made with a large language model as a tool under the author's direction and instructions, as the declaration states; every decision is the author's.

The paper of the engine before 2026-09-25, with its readings, is history, at the commit `cc8b22b8` of the branch `paper-48`.